Quantitative Research

Part 4: Regression

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The lecture at a glance

	Part	Content
	1 – Descriptive Statistics	Basic concepts, levels of measurement, frequencies, measures of central tendency and dispersion
	2 – Graphical Representations	Empirical CDF, histogram, boxplot
	3 – Statistical Inference	Confidence intervals, hypothesis tests, p-values, one-sided tests, types of error
▶	**4 – Regression**	**Correlation, simple and multiple linear regression, inference for regression, diagnostics**
	5 – Classification	Logistic regression, classification rules, confusion matrix, ROC/AUC, Bayes classifiers

All analyses are carried out in **R** (with RStudio); every deck comes with an R script that reproduces its numbers.

Contents – Part 4

1. Relationship Between Two Metric Variables
2. Simple Linear Regression
3. Goodness of Fit: the Coefficient of Determination
4. Residual Analysis
5. Inference in Simple Regression
6. Multiple Linear Regression
7. Diagnostics in Multiple Regression
8. Summary

The common thread of this part

One question runs through every section

How does a variable Y **depend** on one or several variables X – and how *certain* are we about this dependence?

Three steps:

- **Describe** – scatter plot, covariance, correlation, regression line (R^2).
- **Infer** – standard errors, t-tests, confidence intervals and p-values for the coefficients (the tools of Part 3).
- **Extend** – several explanatory variables at once: multiple regression.

Running stories:

The lamp

A new lamp is sold at different prices – how does the sales quantity react to the price? ($n = 9$, simple regression)

The advertising budget

Sales in 200 markets and the TV, radio and newspaper budgets ($n = 200$, multiple regression)

Learning objectives – Part 4

After this part you will be able to ...

- compute and interpret **covariance** and **correlation**,
- estimate a **simple linear regression** by least squares – by hand and in `R`,
- assess the fit with R^2, the **residual plot** and the **QQ-plot**,
- carry out t-**tests**, compute **confidence intervals** and p-**values** for regression coefficients,
- distinguish a **confidence interval** for the mean from a **prediction interval** for a single value,
- interpret the coefficients of a **multiple regression** ("holding the others fixed"),
- use the F-**test**, adjusted R^2, dummy variables and interactions,
- recognise **confounding** and **collinearity**.

Key takeaway

Common thread: a regression coefficient is an *estimate* – report it together with its uncertainty.

List of symbols (1/2)

Symbol	Meaning	Symbol	Meaning
X, Y	Explanatory / dependent variable	(x_i, y_i)	Observation pair i
s_{xy}	Covariance of X and Y	r_{xy}	Correlation coefficient (Pearson)
β_0, β_1	Regression coefficients (true)	$\hat{\beta}_0$, $\hat{\beta}_1$	Estimated coefficients
u_i	Error term	$\hat{u}_i$	Residual $y_i - \hat{y}_i$
$\hat{y}_i$	Fitted value	h_i	Leverage value
S_{xx}, S_{yy}, S_{xy}	Sums of squares / cross-products	TSS, ESS, RSS	Total, explained, residual sum of squares
R^2	Coefficient of determination	$\bar{R}^2$	Adjusted R^2

List of symbols (2/2)

Symbol	Meaning	Symbol	Meaning
σ^2	Variance of the error term	$\hat{\sigma}$	Residual standard error
$\mathrm{SE}(\hat{\beta}_j)$	Standard error of $\hat{\beta}_j$	T	t-statistic of a coefficient
$t_{1-\alpha/2,\,n-2}$	Two-sided quantile, t_{n-2}	$F_{t,n-2}$	CDF of t_{n-2}
p	Number of explanatory variables	$n-p-1$	Degrees of freedom (residuals)
x_{ij}	Value of variable j for obs. i	F	F-statistic
D	Dummy variable (0/1)	VIF_j	Variance inflation factor

As in Part 3: H_0, H_1, α, p-value and confidence level $1-\alpha$. (Context distinguishes the p-value from the number of explanatory variables p.)

Overview

1 Relationship Between Two Metric Variables
- Example and scatter plot
- Covariance and correlation

2 Simple Linear Regression

3 Goodness of Fit: the Coefficient of Determination

4 Residual Analysis

5 Inference in Simple Regression

6 Multiple Linear Regression

7 Diagnostics in Multiple Regression

8 Summary

Relationship between two metric variables

Research question

The aim is to investigate the relationship between two metrically scaled variables.

- **Exogenous variable** X: given from outside, assumed to be independent – also called the *explanatory variable*.
- **Endogenous variable** Y: dependent variable – also called the *variable to be explained*.

Regression – an example

Scenario

A new lamp is offered for testing in several stores at different prices. The price (in Euro) and the sales quantity are observed.

Price x_i (Euro)	80	85	90	95	100	105	110	115	120
Quantity y_i	20	17	15	9	11	8	4	4	2

Assignment of the variables

The price was fixed in every store and not negotiable. Therefore

- the **price** is the *exogenous* variable X,
- the **quantity** is the *endogenous* variable Y (result of buying behaviour, depending on price).

Price-sales and demand function

Price-sales function

The functional relationship $Y = f(X)$ we are looking for is called the *price-sales function*. It answers the question: *What quantity is sold at a given price?*

Demand function

The inverse function $X = f^{-1}(Y)$ is called the *demand function*. It answers the reverse question: *At what price can a fixed quantity be sold?*

In words: Regression always has a *direction*: we explain Y by X, not the other way round.

Regression – a warning

Price x_i (Euro)	80	85	90	95	100	105	110	115	120
Quantity y_i	20	17	15	9	11	8	4	4	2

Warning

- **Do not sort!**
  The values x_i and y_i must *not* be sorted independently of each other (unlike the quantiles of Part 1).
- **Pairs belong together.**
  Each pair (x_i, y_i) comes from *the same observation* (here: from the same store) – the link price $\leftrightarrow$ quantity must not be broken.

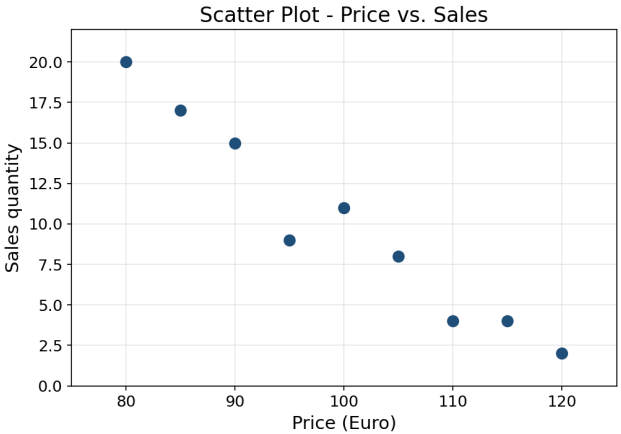
Scatter plot

Idea

A *scatter plot* gives a first visual overview of the relationship between two metric variables X and Y.

- Each observation (x_i, y_i) is plotted as **a single point**.
- The **exogenous value** x_i (here: price) is on the x-axis.
- The **endogenous value** y_i (here: quantity) is on the y-axis.
- From the cloud of points one can read off the *direction*, *strength* and *form* of the relationship.

Scatter plot – lamp example



Covariance

Idea

The *covariance* s_{xy} measures the **joint variability** of two metric variables X and Y.

Computation (sample covariance)

$$s_{xy} = \mathrm{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

with range $-\infty < s_{xy} < +\infty$.

In words: Same divisor $n - 1$ as the sample variance of Part 1 – in fact $s_{xx} = s_x^2$: the covariance of a variable with itself is its variance.

Covariance – interpretation

Sign indicates the direction of the relationship

$s_{xy} > 0$ **Positive (concordant) relationship.** Larger values of X tend to go together with larger values of Y.

$s_{xy} < 0$ **Negative (discordant) relationship.** Larger values of X tend to go together with smaller values of Y.

$s_{xy} = 0$ **No linear relationship.** The cloud of points shows no recognisable linear structure.

Magnitude not directly interpretable

The covariance s_{xy} has the unit "unit of X · unit of Y" and depends on the scaling of the data.

Correlation coefficient

Idea

The correlation coefficient r_{xy} is a **normalised covariance**: it measures the strength and direction of the *linear* relationship between X and Y – *independent of the units of measurement*.

Computation

$$r_{xy} = \frac{s_{xy}}{s_x\, s_y} = \mathrm{Cov}\left(\frac{X-\bar{x}}{s_x}, \frac{Y-\bar{y}}{s_y}\right) \qquad -1 \leq r_{xy} \leq +1.$$

Reading

r_{xy} is the covariance of the *z-standardised* variables. Dividing by $s_x s_y$ cancels out the units.

Correlation coefficient – strength of the relationship

What r_{xy} measures

The correlation coefficient measures the **degree of linear dependence** between X and Y: $r_{xy} = \pm 1$ means that all points lie exactly on a straight line.

Strength of the relationship (rule of thumb)

Range	Interpretation
$\lvert r_{xy} \rvert > 0.8$	strong correlation
$0.5 \leq \lvert r_{xy} \rvert \leq 0.8$	moderate correlation
$\lvert r_{xy} \rvert < 0.5$	weak correlation

Covariance and correlation – lamp example

x_i	80	85	90	95	100	105	110	115	120	$\bar{x} = 100$
y_i	20	17	15	9	11	8	4	4	2	$\bar{y} = 10$
$x_i - \bar{x}$	-20	-15	-10	-5	0	5	10	15	20	
$y_i - \bar{y}$	10	7	5	-1	1	-2	-6	-6	-8	
$(x_i - \bar{x})(y_i - \bar{y})$	-200	-105	-50	5	0	-10	-60	-90	-160	$\Sigma = -670$

Calculation

$$s_{xy} = \frac{-670}{8} = -83.75, \qquad s_x = \sqrt{\frac{1500}{8}} = 13.6931, \qquad s_y = \sqrt{\frac{316}{8}} = 6.2849$$

$$r_{xy} = \frac{-83.75}{13.6931 \cdot 6.2849} = \mathbf{-0.9732}$$

In words: A strong negative linear relationship: the higher the price, the fewer lamps are sold.

Correlation coefficient – limitations

No causal relationship

From r_{xy} alone one *cannot* infer which variable is independent (X) and which is dependent (Y). Correlation is **symmetric**: $r_{xy} = r_{yx}$.

Spurious correlation possible

r_{xy} can be $\neq 0$ **without** a substantive relationship between X and Y – e.g. when both variables are influenced by a common third variable (we return to this as *confounding* in multiple regression).

Remember

A high value of $|r_{xy}|$ is a *statistical finding*, not a *causal statement*.

Covariance and correlation with R

Data

```
price    ← c(80, 85, 90, 95, 100, 105, 110, 115, 120)
quantity ← c(20, 17, 15,  9,  11,   8,   4,   4,   2)
plot(price, quantity)              # scatter plot
```

Covariance and correlation

```
cov(price, quantity)               # -83.75
cor(price, quantity)               # -0.9732 (Pearson is the default)
```

Overview

1 Relationship Between Two Metric Variables

2 Simple Linear Regression
- Least squares estimation
- Exercise
- Leverage values

3 Goodness of Fit: the Coefficient of Determination

4 Residual Analysis

5 Inference in Simple Regression

6 Multiple Linear Regression

7 Diagnostics in Multiple Regression

8 Summary

Motivation

The question

How can the **best-fitting line** through the cloud of points be determined?

- The correlation tells us *how strong* the linear relationship is – but not *by how much* Y changes when X changes.
- A straight line answers exactly this: its **slope** is the change in Y per unit of X.
- Two questions follow: *which* line do we pick – and how *certain* can we be about it (Section "Inference")?

The best line through the cloud of points

Regression line

The linear relationship between X and Y that we are looking for is described by a straight line:

$$y = \beta_0 + \beta_1 x$$

- β_0 *Intercept* – value of y at $x = 0$; without β_0 all lines would pass through the origin.
- β_1 *Slope* – change in y when x increases by one unit.

Estimation and naming

The coefficients β_0 and β_1 are **estimated** from the data. Since they appear only at power 1, one speaks of **linear regression**.

The regression model

Regression model with error term

Since the observations generally do not lie exactly on the line, *deviations* occur:

$$y_i = \underbrace{\beta_0 + \beta_1 x_i}_{\text{systematic part}} + \underbrace{u_i}_{\text{error term}}, \qquad i = 1, \ldots, n.$$

$\beta_0 + \beta_1 x_i$ the share of y_i explained by x_i,

u_i the part of the i-th observation that is *not* explained by X.

Requirement for the error term

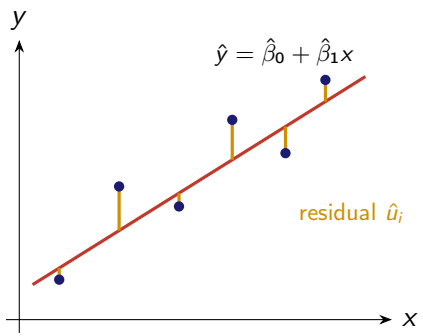
The u_i should contain **no systematic information** about Y any more – everything that can be explained must already be in the linear part.

Intuition: minimise the vertical mistakes

Each line makes a mistake on each point. For observation i, the mistake is the vertical distance between the point y_i and the line above x_i: the **residual** $\hat{u}_i = y_i - \hat{y}_i$.

We pick the line that makes the total squared mistake as small as possible.

In words: Slide the line until the sum of the squared orange segments is minimal.



Method of least squares (OLS)

Idea

We look for the line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ such that the sum of the *squared* residuals is as small as possible (*ordinary least squares*, OLS).

Minimisation problem

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{n} \hat{u}_i^2 = \min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2$$

with residuals $\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$.

Why squared?

- Positive and negative deviations *do not* cancel out (as for the variance in Part 1).
- Large deviations are penalised more heavily than small ones.

Derivation (1/2): set the derivatives to zero

The sum of squared residuals $\text{RSS}(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$ is a "bowl" in (β_0, β_1); at its minimum both partial derivatives vanish:

$$\frac{\partial\, \text{RSS}}{\partial \beta_0} = -2 \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) \stackrel{!}{=} 0 \qquad \triangleright \text{ divide by } -2n$$

$$\implies \quad \bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} \quad \iff \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

In words: The first condition already shows that the line passes through the **centroid** $(\bar{x}, \bar{y})$ of the cloud of points – and that the residuals sum to zero.

Derivation (2/2): the slope

$$\frac{\partial \text{ RSS}}{\partial \beta_1} = -2 \sum_{i=1}^{n} x_i \left(y_i - \beta_0 - \beta_1 x_i\right) \stackrel{!}{=} 0 \qquad \triangleright \text{ insert } \hat{\beta}_0$$

$$0 = \sum_{i=1}^{n} x_i \left[(y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})\right] \qquad \triangleright \text{ use } \textstyle\sum_i (x_i - \bar{x}) = 0$$

$$0 = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) - \hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})^2$$

Key takeaway

Least-squares estimators:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}} = \frac{s_{xy}}{s_x^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Least squares solution

Slope

$$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2}$$

The ratio of the *covariance* of X and Y to the *variance* of X.

Intercept

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Chosen so that the regression line passes through the *centroid* $(\bar{x}, \bar{y})$ of the cloud of points.

In words: Estimated values are marked with a "hat": $\hat{y}_i$, $\hat{\beta}_0$, $\hat{\beta}_1$, $\hat{u}_i$.

Least squares solution – interpretation

Properties of the slope $\hat{\beta}_1$

- $\hat{\beta}_1$ inherits the **sign** of the covariance s_{xy}.
- The slope is exactly 0 if and only if X and Y are **uncorrelated** ($s_{xy} = 0$).
- Connection to the correlation coefficient:

$$\hat{\beta}_1 = r_{xy} \cdot \frac{s_y}{s_x}$$

Remember

Regression, correlation and covariance are closely related: the signs agree, and the slope differs from r_{xy} only by the ratio of standard deviations s_y/s_x.

Least squares – lamp example (calculation)

From the table of the correlation example: $\bar{x} = 100$, $\bar{y} = 10$, $S_{xy} = -670$, $S_{xx} = 1500$.

Calculation

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-670}{1500} = \mathbf{-0.4467} \qquad \left(= \frac{s_{xy}}{s_x^2} = \frac{-83.75}{187.5} \right)$$

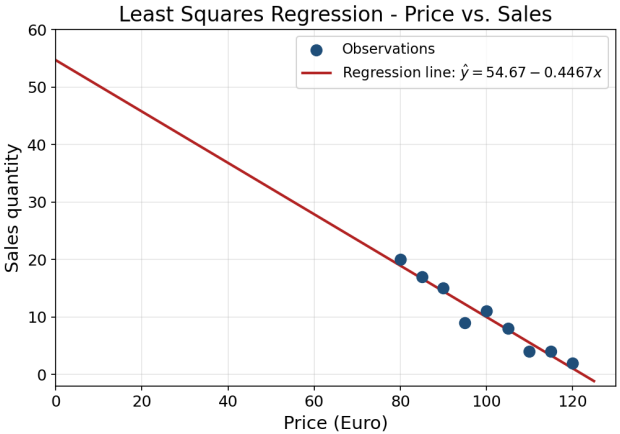
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 10 - (-0.4467) \cdot 100 = 10 + 44.67 = \mathbf{54.67}$$

Estimated regression line

$$\hat{y}_i = 54.67 - 0.4467\, x_i$$

In words: Check: $\hat{\beta}_1 = r_{xy}\, s_y / s_x = -0.9732 \cdot 6.2849 / 13.6931 = -0.4467$ ✓

Least squares – lamp example (graph)



Example: interpretation of the slope

Estimated regression line

$$\hat{y}_i = 54.67 - 0.4467 \cdot x_i$$

Slope $\hat{\beta}_1 = -0.4467$

If the price increases by 1 Euro, the sales quantity decreases *on average* by 0.4467 units – or: a price increase of 10 Euro costs about 4.5 lamps.

Economic interpretation

The negative sign confirms the expected *inverse* relationship – a classical downward-sloping **price-sales function**.

Example: intercept $\hat{\beta}_0 = 54.67$

Formal meaning

Intersection of the regression line with the y-axis – at a price of 0 Euro one would compute sales of 54.67 lamps.

Not meaningful in substantive terms

The value $x = 0$ lies far outside the observed price range (80 to 120 Euro). The regression line is only reliable within the range of the data; outside the observed x-values (*extrapolation*) extreme caution is required.

Key takeaway

Interpret the slope as an average change *within* the data range – and the intercept only if $x = 0$ is a meaningful value that lies near the data.

Linear regression with R

Coefficients by hand

```
(beta1 ← cov(price, quantity) / var(price))       # -0.4467
(beta0 ← mean(quantity) - beta1 * mean(price))    # 54.6667
beta1 * sd(price) / sd(quantity)                  # = cor(price, quantity)
```

Regression with `lm()` (linear model)

```
reg ← lm(quantity ~ price)        # read: quantity explained by price
coef(reg)                         # 54.6667  -0.4467
plot(price, quantity, xlim = c(0, 125), ylim = c(0, 60))
abline(reg)
```

Exercise – least squares by hand

Problem statement

Five observations are given:

$$(x_i, y_i) = (1,3),\ (2,4),\ (3,2),\ (4,5),\ (5,6).$$

1. Compute the means $\bar{x}$ and $\bar{y}$.
2. Compute $S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})$ and $S_{xx} = \sum(x_i - \bar{x})^2$, then $\hat{\beta}_1$ and $\hat{\beta}_0$.
3. Write down the fitted line and predict y at $x = 6$.
4. Interpret the slope in words.

Solution (1/2) – least squares by hand

(1) Means: $\bar{x} = \frac{1+2+3+4+5}{5} = 3, \quad \bar{y} = \frac{3+4+2+5+6}{5} = 4.$

(2) Deviations and sums:

$x_i - \bar{x}$	-2	-1	0	1	2	
$y_i - \bar{y}$	-1	0	-2	1	2	
$(x_i - \bar{x})(y_i - \bar{y})$	2	0	0	1	4	$S_{xy} = 7$
$(x_i - \bar{x})^2$	4	1	0	1	4	$S_{xx} = 10$

Estimates

$$\hat{\beta}_1 = \frac{7}{10} = \mathbf{0.7}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 4 - 0.7 \cdot 3 = \mathbf{1.9}.$$

Solution (2/2) – least squares by hand

(3) Fitted line and prediction:

$$\hat{y} = 1.9 + 0.7\,x, \qquad \hat{y}(6) = 1.9 + 0.7 \cdot 6 = 6.1.$$

Check: at $x = \bar{x} = 3$, $\hat{y} = 1.9 + 2.1 = 4 = \bar{y}$ ✓ (the line passes through the centroid).

(4) Interpretation: a one-unit increase in x goes along with an *average* increase of 0.7 units in y.

Careful

$x = 6$ lies outside the observed range $[1, 5]$ – the prediction 6.1 is an (mild) **extrapolation** and less trustworthy than a prediction inside the data range.

Method of least squares – leverage values

Disadvantage of the method

Single points lying far away from the centre of the data cloud (e.g. measurement errors or *outliers*) can exert a strong influence on the regression line – the deviations enter the estimation *squared*.

Leverage value h_i

The leverage value measures how **unusual** the x-value of observation i is – and thus how strongly this point can pull the line towards itself:

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^{n}(x_j - \bar{x})^2}, \qquad \frac{1}{n} \leq h_i \leq 1.$$

In words: The further x_i lies from the mean $\bar{x}$, the larger h_i becomes (a long "lever arm").

Leverage values – rule of thumb

Rule of thumb for points with high leverage

Simple regression Values with $h_i > \frac{4}{n}$ are **flagged** as high-leverage points.

Small samples Stricter threshold $h_i > \frac{6}{n}$.

(In multiple regression with p explanatory variables: $h_i > 2(p+1)/n$ – see the section "Diagnostics".)

Caution

A flagged point is a reason to **inspect** it (measurement error? special case?) and to compare the fit with and without it – not to delete it automatically. Every removal changes the estimate and must be justified *substantively*.

Leverage values – lamp example

$n = 9$, $\bar{x} = 100$, $S_{xx} = 1500$:

x_i	80	85	90	95	100	105	110	115	120
h_i	0.378	0.261	0.178	0.128	0.111	0.128	0.178	0.261	0.378

Calculation for the first store

$$h_1 = \frac{1}{9} + \frac{(80 - 100)^2}{1500} = 0.1111 + 0.2667 = 0.378$$

Assessment

Threshold $4/n = 0.444$: no store has a high leverage – the prices are spread evenly; the extreme prices 80 and 120 have the largest (but still harmless) influence.

Overview

1 Relationship Between Two Metric Variables

2 Simple Linear Regression

3 Goodness of Fit: the Coefficient of Determination
- Variance decomposition
- Coefficient of determination
- Useful transformations
- Exercise

4 Residual Analysis

5 Inference in Simple Regression

6 Multiple Linear Regression

7 Diagnostics in Multiple Regression

8 Summary

Variance decomposition

Idea

The regression decomposes the variance s_y^2 into a variance *on* the regression line $s_{\hat{y}}^2$ and a variance *around* the regression line $s_{\hat{u}}^2$ (the **residual variance**).

Variance decomposition

$$s_y^2 = s_{\hat{y}}^2 + s_{\hat{u}}^2$$

This decomposition is also called **analysis of variance (ANOVA)**.

In words

(total variance of y) = (variance explained by the regression) + (residual variance)

Variance decomposition – sums of squares

Sum-of-squares decomposition

$$\underbrace{\sum_{i=1}^{n}(y_i-\bar{y})^2}_{\text{TSS (total)}} = \underbrace{\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2}_{\text{ESS (explained)}} + \underbrace{\sum_{i=1}^{n}\hat{u}_i^2}_{\text{RSS (residual)}}$$

In variances

Dividing all three sums by $n-1$ gives the decomposition of variances $s_y^2 = s_{\hat{y}}^2 + s_{\hat{u}}^2$ (the fitted values have mean $\bar{y}$, the residuals mean 0):

- s_y^2 total variance of y,
- $s_{\hat{y}}^2$ variance *explained* by the regression,
- $s_{\hat{u}}^2$ *residual variance* (unexplained part).

Variance decomposition – link to the method of least squares

Starting point

The variance *around* the regression line is the part of the total variance that cannot be explained by the regression. The coefficients are computed by **minimising exactly this residual variance** (least squares). Therefore it is **minimal**.

Consequence for the variance decomposition

$$\underbrace{s_y^2}_{\text{fixed}} = \underbrace{s_{\hat{y}}^2}_{\text{maximal}} + \underbrace{s_{\hat{u}}^2}_{\text{minimal}}$$

$\Rightarrow$ No other straight line explains a larger share of the variance of y than the least-squares line.

Coefficient of determination

Definition

The **coefficient of determination** R^2 is the ratio of the variance on the regression line to the total variance:

$$R^2 = \frac{s_{\hat{y}}^2}{s_y^2} = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} \qquad 0 \leq R^2 \leq 1.$$

Interpretation

The coefficient of determination gives the *share* of the total variance of y that is explained by the regression.

Coefficient of determination – extreme cases

$R^2 = 1$: perfect fit

The regression explains the *entire* variance of y. **All points lie exactly on the regression line**; all residuals are zero: $s^2_{\hat{u}} = 0$.

$R^2 = 0$: no explained variance

The regression line explains no variance: $s^2_{\hat{y}} = 0$. **The regression line has zero slope** ($\hat{\beta}_1 = 0$) and runs at the level of the mean $\bar{y}$.

Rule of thumb

The closer R^2 is to 1, the better the linear model explains the data. A low R^2 does *not* mean that X and Y are independent – only that the *linear* relationship is weak.

Explained variance – derivation

Step 1: mean of the fitted values

Since $\hat{\beta}_0$ and $\hat{\beta}_1$ are constants: $\bar{\hat{y}} = \frac{1}{n}\sum_i \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1\,\bar{x}\;(= \bar{y})$.

Step 2: form the difference – the intercept cancels out

$$\hat{y}_i - \bar{\hat{y}} = \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right) - \left(\hat{\beta}_0 + \hat{\beta}_1 \bar{x}\right) = \hat{\beta}_1\,(x_i - \bar{x}).$$

Step 3: square, sum, divide by $n - 1$

$$s^2_{\hat{y}} = \tfrac{1}{n-1}\sum_{i=1}^{n} \hat{\beta}_1^2 (x_i - \bar{x})^2 = \hat{\beta}_1^2 \cdot \underbrace{\tfrac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}_{= s_x^2} = \hat{\beta}_1^2\, s_x^2.$$

In words: The explained variance is the variance of x, scaled by the **square** of the slope.

Coefficient of determination in simple regression

Alternative formula

Substituting $s_{\hat{y}}^2 = \hat{\beta}_1^2 \, s_x^2$ into $R^2 = s_{\hat{y}}^2 / s_y^2$:

$$R^2 \;=\; \hat{\beta}_1^2 \, \frac{s_x^2}{s_y^2} \;=\; \left(\hat{\beta}_1 \frac{s_x}{s_y} \right)^2 \;=\; \left(\frac{s_{xy}}{s_x \, s_y} \right)^2 \;=\; r_{xy}^2 .$$

Root relation

$\sqrt{R^2} = |r_{xy}|$: the square root of R^2 is the **absolute value** of the correlation coefficient – the sign (direction) is lost and must be read from $\hat{\beta}_1$ or r_{xy}.

Only valid in simple regression

With several explanatory variables, R^2 is no longer the square of a single correlation coefficient.

Coefficient of determination – lamp example

Via the sums of squares

$$\text{TSS} = \sum (y_i - \bar{y})^2 = 316, \qquad \text{RSS} = \sum \hat{u}_i^2 = 16.733, \qquad \text{ESS} = 316 - 16.733 = 299.267$$

$$R^2 = 1 - \frac{16.733}{316} = \frac{299.267}{316} = \mathbf{0.9470}$$

Cross-checks

$$R^2 = \hat{\beta}_1^2 \frac{s_x^2}{s_y^2} = (-0.4467)^2 \cdot \frac{187.5}{39.5} = 0.9470, \qquad R^2 = r_{xy}^2 = (-0.9732)^2 = 0.9470 \checkmark$$

In words: About 95 % of the variation of the sales quantity across the stores is explained by the price.

Coefficient of determination – assessment

In words: Because $R^2 = r_{xy}^2$, a correlation that looks high gives a much smaller R^2: $r_{xy} = 0.7$ yields only $R^2 = 0.49$.

Rule of thumb (used in this course)

Value of R^2	Assessment
$R^2 < 0.6$	weak fit
$0.6 \leq R^2 < 0.8$	moderate fit
$R^2 \geq 0.8$	good fit

Context matters

What counts as "good" depends on the field: in finance or the social sciences an R^2 of 0.1 can be valuable, in physics 0.95 may be poor.

Common misinterpretations of R^2

A high R^2 does *not* prove

- that the model is *correct* (a curved relationship can still give a high R^2),
- that X *causes* Y,
- that predictions *outside* the data range are good.

A low R^2 does *not* prove

- that the model is *useless*,
- that X is *irrelevant* – the slope can still be clearly different from zero (see the t-test later),
- that there is no *other* relationship.

Key takeaway

Never judge a regression by R^2 alone – always look at the residual plot and at the uncertainty of the coefficients.

Useful transformations of the formulas

Goal

The relationships between **correlation coefficient**, **slope** $\hat{\beta}_1$ and **coefficient of determination** R^2 greatly simplify calculations by hand – in particular because the factors $\frac{1}{n-1}$ *cancel*.

Overview

$$r_{xy} = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}, \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad R^2 = r_{xy}^2 = \frac{S_{xy}^2}{S_{xx}\,S_{yy}}$$

with the plain sums $S_{xx} = \sum(x_i - \bar{x})^2$, $S_{yy} = \sum(y_i - \bar{y})^2$, $S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})$.

Correlation coefficient – cancelling the prefactor

Starting formula

$$r_{xy} = \frac{s_{xy}}{s_x\, s_y} = \frac{\frac{1}{n-1} S_{xy}}{\sqrt{\frac{1}{n-1} S_{xx}} \cdot \sqrt{\frac{1}{n-1} S_{yy}}}$$

Combining the roots

$$\sqrt{\tfrac{1}{n-1} A} \cdot \sqrt{\tfrac{1}{n-1} B} = \tfrac{1}{n-1} \sqrt{A\, B} \quad \Longrightarrow \quad r_{xy} = \frac{S_{xy}}{\sqrt{S_{xx}\, S_{yy}}}.$$

Remember

The factor $\frac{1}{n-1}$ appears in numerator *and* denominator – it can be omitted. The same holds for $\hat{\beta}_1 = s_{xy}/s_x^2 = S_{xy}/S_{xx}$.

Coefficient of determination – via the residuals

Three equivalent formulas

$$R^2 = \frac{\sum_i (\hat{y}_i - \bar{y})^2}{\sum_i (y_i - \bar{y})^2} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} = r_{xy}^2 \quad \text{(simple regression)}$$

In words

$$R^2 = 1 - \frac{\text{sum of squared residuals}}{\text{sum of squared deviations of } y \text{ from the mean}}$$

In words: The middle formula is the one that also works in multiple regression – it is the one R uses.

Coefficient of determination in R

R command

```
summary(reg)                    # full output, incl. "Multiple R-squared"
summary(reg)$r.squared          # 0.9470
```

Manual variants

```
RSS ← sum(resid(reg)^2)                          # 16.7333
TSS ← sum((quantity - mean(quantity))^2)         # 316
1 - RSS / TSS                                    # 0.9470
cor(price, quantity)^2                           # simple regression only
```

Exercise – R^2 and the residual sum of squares

Problem statement

For the five points of the exercise "least squares by hand" ($\hat{y} = 1.9 + 0.7x$) we have $S_{xx} = 10$, $S_{xy} = 7$ and $S_{yy} = \text{TSS} = 10$.

1. Compute the correlation coefficient r_{xy}.
2. Compute the fitted values, the residuals and RSS.
3. Compute R^2 in two ways and interpret it.

Solution (1/2) – R^2 and the residual sum of squares

(1) Correlation: $r_{xy} = \dfrac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} = \dfrac{7}{\sqrt{10 \cdot 10}} = \mathbf{0.7}$.

(2) Fitted values and residuals ($\hat{y}_i = 1.9 + 0.7 x_i$):

y_i	3	4	2	5	6	
$\hat{y}_i$	2.6	3.3	4.0	4.7	5.4	
$\hat{u}_i$	0.4	0.7	-2.0	0.3	0.6	$\Sigma = 0$ ✓
$\hat{u}_i^2$	0.16	0.49	4.00	0.09	0.36	RSS $= 5.1$

In words: The residuals sum to zero – a property of every least-squares line with intercept.

Solution (2/2) – R^2 and the residual sum of squares

(3) Two ways to R^2:

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{5.1}{10} = \mathbf{0.49}, \qquad R^2 = r_{xy}^2 = 0.7^2 = 0.49 \checkmark$$

Interpretation

49 % of the variance of y is explained by x, 51 % remain unexplained: a **weak** fit – although $r_{xy} = 0.7$ looked "moderate".

Common mistake

Reading $R^2 = 0.49$ as "49 % of the points are predicted correctly". R^2 is a share of *variance*, not a hit rate.

Overview

1 Relationship Between Two Metric Variables

2 Simple Linear Regression

3 Goodness of Fit: the Coefficient of Determination

4 Residual Analysis
- Assumptions and residual plot
- QQ-plot
- Exercises

5 Inference in Simple Regression

6 Multiple Linear Regression

7 Diagnostics in Multiple Regression

8 Summary

Assumptions on the error term

Four standard assumptions

Expected value 0 $E(u_i) = 0$ – no systematic error on average.

Homoscedasticity $\mathrm{Var}(u_i) = \sigma^2$ for all i – *constant* variance, independent of x_i.

No autocorrelation $\mathrm{Cov}(u_i, u_j) = 0$ for $i \neq j$ – the errors are pairwise uncorrelated.

Normal distribution $u_i \sim N(0, \sigma^2)$ – required for confidence intervals and hypothesis tests (Section "Inference").

Important

The assumptions are **not self-evident** – they have to be **checked**. We check them with the *residuals* $\hat{u}_i$, our observable stand-ins for the unobservable errors u_i.

Residuals – required structure

Basic principle

The residuals $\hat{u}_i$ should contain **no systematic structure**: in the residual plot no recognisable pattern should be visible. Any structure indicates that the model has *not* explained something it should have explained.

Typical structures pointing to model errors

Systematic patterns – indicate a **wrong model equation** (e.g. non-linear relationship, missing variable).

Autocorrelation – successive residuals are correlated ($\text{Cov}(u_i, u_j) \neq 0$); typical for time series.

Heteroscedasticity – the spread of the residuals depends on x_i ($\text{Var}(u_i)$ not constant).

Residual plot

Idea

A *residual plot* displays the estimated residuals graphically in order to check the **structure of the deviations** from the regression line.

Estimated residuals

$$\hat{u}_i \;=\; y_i - \hat{y}_i \;=\; y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)$$

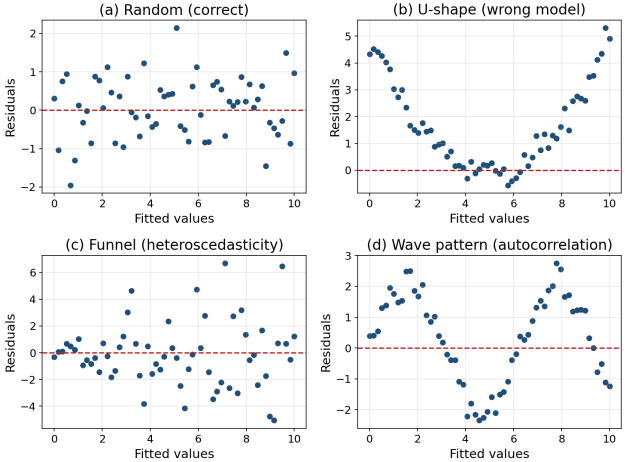
Layout of the plot

x-**axis:** fitted values $\hat{y}_i$ y-**axis:** residuals $\hat{u}_i$ (points $(\hat{y}_i,\ \hat{u}_i)$, $i = 1, \ldots, n$)

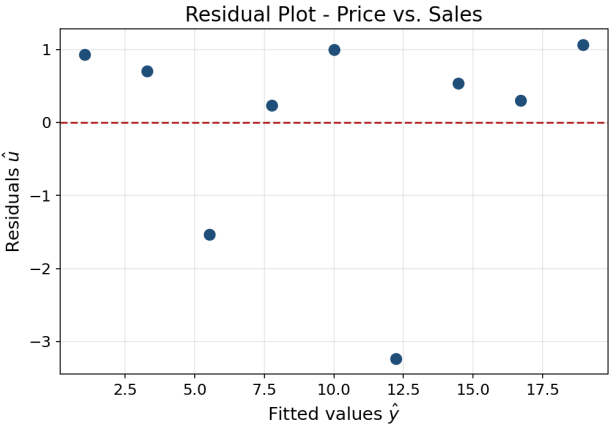
What we want to see

A *structureless* cloud of points around the zero line.

Residual plot – pattern recognition



Residual plot – lamp example



Observation

No systematic pattern visible – the residuals scatter around zero without curvature or funnel shape.

But

One store ($x = 95$, $\hat{u}_4 = -3.23$) deviates clearly – with $n = 9$, single points weigh heavily.

Creating a residual plot with R

R commands

```
reg ← lm(quantity ∼ price)
plot(fitted(reg), resid(reg), xlab = "fitted values",
     ylab = "residuals", main = "Residual plot")
abline(h = 0, lty = 2)          # zero line for orientation
```

Helper functions

- `fitted(reg)`: fitted values $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ (x-axis)
- `resid(reg)`: residuals $\hat{u}_i = y_i - \hat{y}_i$ (y-axis)

QQ-plot

Assumption

The errors of the linear regression are **normally distributed**.

Tool for checking

Checking normality with the help of a **QQ-plot** (quantile-quantile plot).

In words: The normal distribution is the basis of the t-tests and confidence intervals for the coefficients – exactly as in Part 3 for the mean.

What is a QQ-plot?

Definition

The *QQ-plot* is a **quantile-quantile graph**: the quantiles of two distributions are plotted against each other on the two axes.

Layout and reference line

For better comparison a **line connecting the** 25 % **and the** 75 % **quantile** is drawn.

Interpretation

The further the points lie from the line, **the more dissimilar** the two distributions are. If all points lie on the line, the distributions have the same shape.

What is a QQ-plot? – example

Lamp example

x (price)	80	85	90	95	100	105	110	115	120
y (sales)	20	17	15	9	11	8	4	4	2

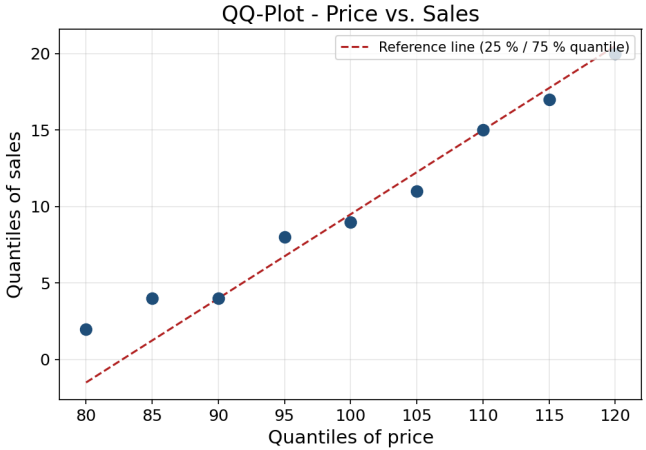
Step 1 – sort the data

For a QQ-plot, the x and y data are **sorted separately** (here – and only here – the pairs are broken up, because we compare two *distributions*):

x^{sort}	80	85	90	95	100	105	110	115	120
y^{sort}	2	4	4	8	9	11	15	17	20

The pairs $(x_{(i)}, y_{(i)})$ are then plotted.

What is a QQ-plot? – example (graph)



Creating a QQ-plot with R

Drawing the QQ-plot

```
qqplot(price, quantity)
```

Adding a line to the QQ-plot

```
qqline(quantity,
       distribution = function(p) quantile(price, prob = p),
       lty = 2)
```

Idea

`qqplot()` produces a QQ-diagram for the *empirical* distributions of two samples; `qqline()` adds a reference line through the 25 % and 75 % quantiles.

Inspecting the residuals with the QQ-plot

QQ-plot sample vs. theory

The QQ-plot can also compare an empirical distribution with a *theoretical* distribution – here the residuals with the **standard normal distribution**.

Construction

y-axis sorted residuals $\hat{u}_{(i)}$

x-axis quantiles z_i of the standard normal distribution

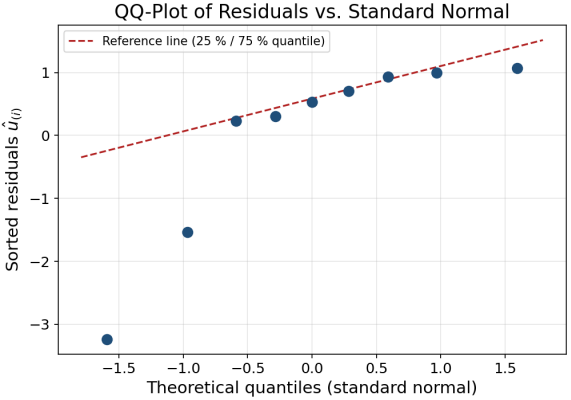
The theoretical quantiles are obtained from

$$\Phi(z_i) = \frac{i - 0.5}{n}, \qquad z_i = \Phi^{-1}\left(\frac{i-0.5}{n}\right).$$

Inspecting the residuals with the QQ-plot – example

Residual $\hat{u}_i$	$\hat{u}_{(i)}$ **sorted**	**Index** i	$\frac{i-0.5}{n}$	**Quantile** z_i
1.0667	−3.2333	1	0.0556	−1.5932
0.3000	−1.5333	2	0.1667	−0.9674
0.5333	0.2333	3	0.2778	−0.5895
−3.2333	0.3000	4	0.3889	−0.2822
1.0000	0.5333	5	0.5000	0.0000
0.2333	0.7000	6	0.6111	0.2822
−1.5333	0.9333	7	0.7222	0.5895
0.7000	1.0000	8	0.8333	0.9674
0.9333	1.0667	9	0.9444	1.5932

Inspecting the residuals with the QQ-plot – interpretation



Observation

In the middle good fit to the normal distribution. **At the lower end** two residuals lie clearly below the line.

Conclusion

$n = 9$ – only very few data points! The normality assumption is **questionable**; tests and confidence intervals for this example should be read with caution.

QQ-plot of the residuals against the normal distribution in R

Create QQ-plot

```
qqnorm(resid(reg))
```

Add line through the quartiles

```
qqline(resid(reg))
```

What the commands do

`qqnorm()` computes the plotting positions $p_i = (i - 0.5)/n$, the normal quantiles $z_i = \Phi^{-1}(p_i)$ and plots the pairs $(z_i, \hat{u}_{(i)})$. `qqline()` adds a reference line through the 25 % and 75 % quantiles.

Exercise – reading residual plots

Problem statement

For each pattern in a residual plot (residuals against fitted values), name the problem and a possible remedy.

1. The residuals form a clear U-shape around zero.
2. The residuals fan out into a funnel: small spread at low fitted values, large spread at high fitted values.
3. The residuals form a random band of roughly constant width around zero.
4. One point lies far away from all others in x-direction and pulls the line towards itself.

Solution – reading residual plots

1. **Non-linearity:** the straight line is the wrong model equation. *Remedy:* add a quadratic term x^2 or transform x (e.g. $\log x$).
2. **Heteroscedasticity:** the variance of the errors is not constant – standard errors and tests become unreliable. *Remedy:* transform y (e.g. $\log y$).
3. **Everything fine:** the assumptions look satisfied – no action needed.
4. **High leverage point:** check h_i against $4/n$; inspect the point and compare the fit with and without it.

Common mistake

Judging the assumptions from R^2. A high R^2 can coexist with patterns 1, 2 and 4 – only the residual plot reveals them.

Exercise – body heights and shoe sizes: overview

Data basis

Body heights k_i (in cm) and shoe sizes s_i of $n = 15$ persons (the body heights are those of the exercise in Part 2):

k_i	164	162	168	198	166	166	192	160	151	175	180	160	168	175	168
s_i	39	38	39	46	40	39	45	37	36	42	43	38	41	40	39

Tasks

Task 1 Compute the **covariance** and the **correlation coefficient**.

Task 2 Estimate the regression $s_i = \beta_0 + \beta_1 k_i + u_i$, draw the scatter plot with regression line and the residual plot, and compute R^2.

Task 3 Create and interpret a **QQ-plot** of the residuals.

Solution Task 1 – means and covariance

Step 1: means

$$\bar{k} = \frac{1}{15}\sum_{i=1}^{15} k_i = \frac{2553}{15} = 170.2, \qquad \bar{s} = \frac{1}{15}\sum_{i=1}^{15} s_i = \frac{602}{15} = 40.1333$$

Step 2: covariance

$$s_{ks} = \frac{1}{n-1}\sum_{i=1}^{n}(k_i - \bar{k})(s_i - \bar{s}) = \frac{471.6}{14} = 33.6857$$

In words: Positive covariance: taller persons tend to have larger shoe sizes.

Solution Task 1 – standard deviations and correlation

Step 3: sample variances / standard deviations

$$s_k^2 = \frac{2122.4}{14} = 151.6 \;\Rightarrow\; s_k = 12.3126, \qquad s_s^2 = \frac{111.7333}{14} = 7.9810 \;\Rightarrow\; s_s = 2.8251$$

Step 4: correlation coefficient

$$r_{ks} = \frac{s_{ks}}{s_k\, s_s} = \frac{33.6857}{12.3126 \cdot 2.8251} = \mathbf{0.9684}$$

Interpretation

Very strong positive linear relationship between body height and shoe size.

Solution Task 2 – regression coefficients

Slope $\hat{\beta}_1$

$$\hat{\beta}_1 = \frac{s_{ks}}{s_k^2} = \frac{33.6857}{151.6} = \frac{471.6}{2122.4} = \mathbf{0.2222}$$

Intercept $\hat{\beta}_0$

$$\hat{\beta}_0 = \bar{s} - \hat{\beta}_1 \, \bar{k} = 40.1333 - 0.2222 \cdot 170.2 = \mathbf{2.3147}$$

Estimated model

$$\hat{s}_i = 2.3147 + 0.2222 \, k_i$$

Per 1 cm of body height the shoe size increases on average by about 0.22 – i.e. by about one size per 4.5 cm.

Solution Task 2 – coefficient of determination R^2

Via the alternative formula

$$R^2 = \hat{\beta}_1^2 \frac{s_k^2}{s_s^2} = (0.2222)^2 \cdot \frac{151.6}{7.9810} = 0.9379$$

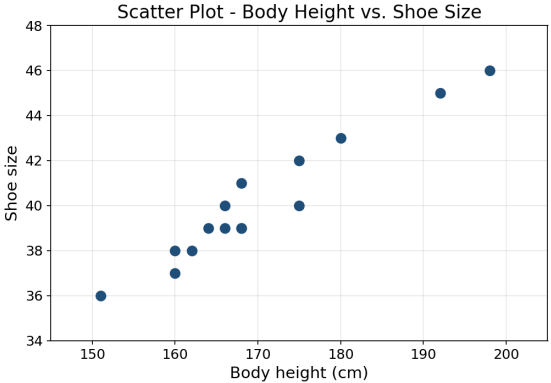
Cross-check via r_{ks}^2

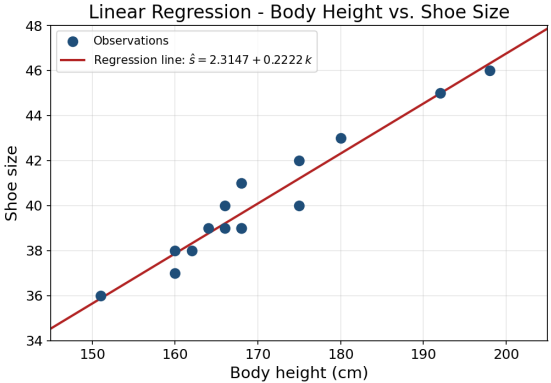
$$R^2 = r_{ks}^2 = (0.9684)^2 = 0.9379 \quad \checkmark$$

Interpretation

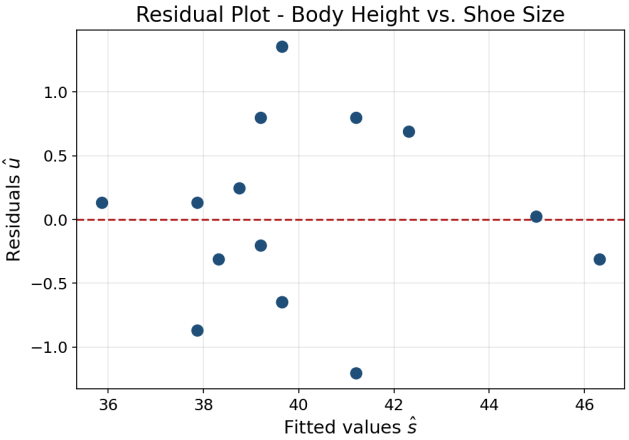
About 93.8 % of the variance of the shoe sizes is explained by body height – $R^2 \geq 0.8$ indicates a **good fit**.

Solution Task 2 – scatter plot with regression line





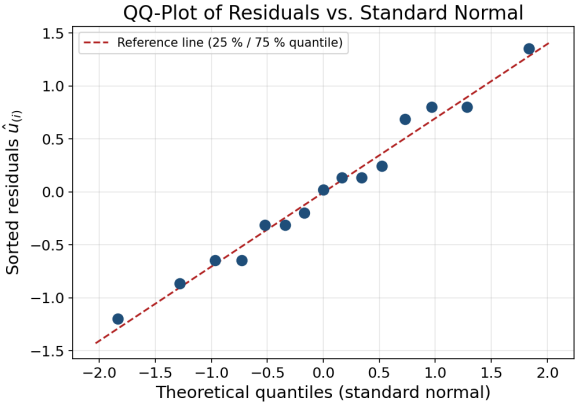
Solution Task 2 – residual plot



Observation

No systematic structure visible ✓ – no curve, no funnel.

Solution Task 3 – QQ-plot of the residuals



Procedure: residuals $\hat{u}_i = s_i - \hat{s}_i$, sorted; normal quantiles $z_i = \Phi^{-1}\left(\frac{i-0.5}{15}\right)$; plot $(z_i, \hat{u}_{(i)})$.

Conclusion

The points lie almost perfectly on the line – **the normality assumption is met**. This model is a good candidate for inference (next section).

Overview

1 Relationship Between Two Metric Variables

2 Simple Linear Regression

3 Goodness of Fit: the Coefficient of Determination

4 Residual Analysis

5 Inference in Simple Regression
- Sampling variability
- t-test for a coefficient
- Confidence intervals
- p-values and R output
- Prediction
- Exercises

6 Multiple Linear Regression

7 Diagnostics in Multiple Regression

8 Summary

Why regression coefficients carry uncertainty

Intuition (as in Part 3)

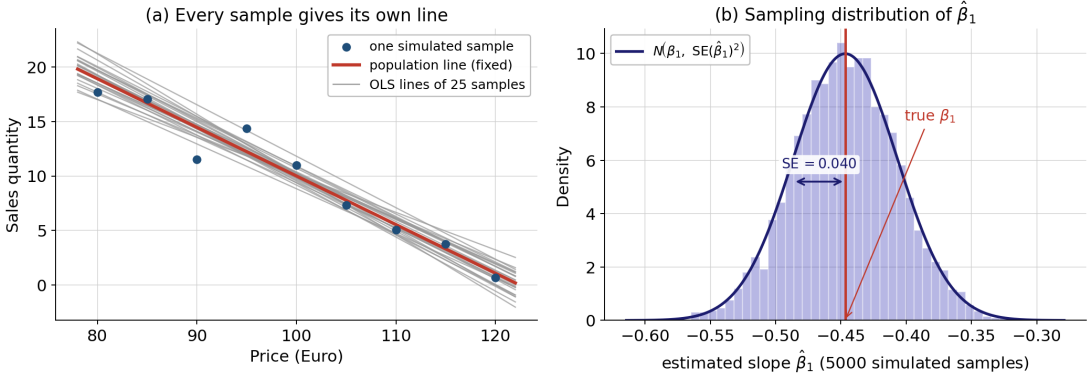
The nine stores are only a **sample**. Other stores – or the same stores in another week – would give slightly different sales, hence a slightly different line.

- The **true** coefficients β_0, β_1 (population line) are fixed but unknown.
- The **estimates** $\hat{\beta}_0, \hat{\beta}_1$ are computed from a random sample – they are **random variables** with a sampling distribution, exactly like $\bar{X}$ in Part 3.
- Questions: How much does $\hat{\beta}_1$ wobble? Is the true slope possibly 0? Which values of β_1 are compatible with the data?

Key takeaway

The tools are the same as for the mean: **standard error**, t-**test**, **confidence interval** and p-**value**.

Population line vs. sample lines



Simulation with the lamp prices: (a) every sample gives its own OLS line around the fixed population line; (b) over 5000 samples the slopes $\hat{\beta}_1$ are normally distributed around β_1 – their standard deviation is the **standard error**.

Sampling distribution of the slope

Result (under the four assumptions on u_i)

$$\hat{\beta}_1 \sim N\left(\beta_1,\ \frac{\sigma^2}{S_{xx}}\right), \qquad \mathrm{SE}(\hat{\beta}_1) = \frac{\sigma}{\sqrt{S_{xx}}} = \frac{\sigma}{\sqrt{\sum_i (x_i - \bar{x})^2}}.$$

- $E(\hat{\beta}_1) = \beta_1$: the estimator is **unbiased** – on average it hits the true slope.
- **Less noise** ($\sigma \downarrow$) $\Rightarrow$ smaller SE.
- **More spread in** x ($S_{xx} \uparrow$) $\Rightarrow$ smaller SE: a long "lever arm" fixes the slope precisely.
- **More observations** increase S_{xx} $\Rightarrow$ smaller SE.

In words: Compare Part 3: $\mathrm{SE}(\bar{X}) = \sigma/\sqrt{n}$. Here the role of n is taken by S_{xx}.

Estimating σ: the residual standard error

Problem

The error variance σ^2 is unknown – as σ in Part 3. We estimate it from the residuals.

Residual standard error

$$\hat{\sigma} = \sqrt{\frac{\text{RSS}}{n-2}} = \sqrt{\frac{1}{n-2}\sum_{i=1}^{n}\hat{u}_i^2}, \qquad \text{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}}, \qquad \text{SE}(\hat{\beta}_0) = \hat{\sigma}\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}.$$

- **Why $n-2$?** Two parameters (β_0, β_1) were estimated – two degrees of freedom are used up (Part 1: one parameter $\bar{x} \Rightarrow n-1$).
- $\hat{\sigma}$ is measured in the **units of** y: the typical size of a residual.

Standard errors – lamp example

Given: $n = 9$, RSS $= 16.733$, $S_{xx} = 1500$, $\bar{x} = 100$.

Calculation

$$\hat{\sigma} = \sqrt{\frac{16.733}{9-2}} = \sqrt{2.3905} = \mathbf{1.5461}$$

$$\mathrm{SE}(\hat{\beta}_1) = \frac{1.5461}{\sqrt{1500}} = \frac{1.5461}{38.730} = \mathbf{0.03992}$$

$$\mathrm{SE}(\hat{\beta}_0) = 1.5461\sqrt{\frac{1}{9} + \frac{100^2}{1500}} = 1.5461 \cdot 2.6034 = \mathbf{4.0252}$$

In words: A typical store deviates by about 1.5 lamps from the line; the slope -0.4467 is uncertain by about ± 0.04 (one standard error).

The first question: is there any relationship at all?

Hypotheses

$$H_0 : \beta_1 = 0 \qquad \text{vs.} \qquad H_1 : \beta_1 \neq 0$$

- H_0: the price has **no** (linear) influence on the sales quantity – the population line is flat.
- H_1: the price has *some* influence.
- As in Part 3: what we want to **establish** goes into H_1.

Key takeaway

Rejecting $H_0 : \beta_1 = 0$ is what licenses us to interpret the slope at all – a slope that could just as well be zero tells us nothing.

The t-statistic – the same recipe as in Part 3

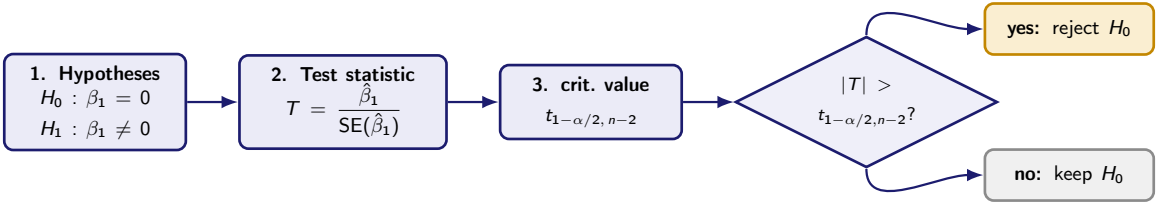
Test statistic

$$T = \frac{\hat{\beta}_1 - \beta_{1,0}}{\text{SE}(\hat{\beta}_1)} \sim t_{n-2} \text{ under } H_0 : \beta_1 = \beta_{1,0} \qquad \text{(usually } \beta_{1,0} = 0\text{)}$$

	Part 3: mean	**Part 4: slope**
Estimate	$\bar{x}$	$\hat{\beta}_1$
Hypothesized value	μ_0	$\beta_{1,0}$ (mostly 0)
Standard error	$s/\sqrt{n}$	$\hat{\sigma}/\sqrt{S_{xx}}$
Distribution under H_0	t_{n-1}	t_{n-2}
Reject (two-sided) if	$\lvert T_n \rvert > t_{1-\alpha/2,\, n-1}$	$\lvert T \rvert > t_{1-\alpha/2,\, n-2}$

In words: "How many standard errors is the estimate away from the hypothesized value?"

The test recipe for a regression coefficient



- Fix α (typically 0.05) *before* looking at the data.
- Equivalent decision via the p-value: reject H_0 if $p < \alpha$, with $p = 2\,F_{t,n-2}(-|T|)$.

t-test – lamp example

Data: $\hat{\beta}_1 = -0.4467$, $\mathrm{SE}(\hat{\beta}_1) = 0.03992$, $n = 9$, $\alpha = 0.05$, $t_{0.975;\,7} = 2.3646$.

$$T = \frac{-0.4467 - 0}{0.03992} = -11.19$$

$$|T| = 11.19 \;>\; 2.3646 = t_{0.975;\,7}$$

Decision

$H_0 : \beta_1 = 0$ is **rejected**. The price has a significant influence on the sales quantity. The p-value is $p = 2\,F_{t,7}(-11.19) = 0.00001$.

In words: The estimated slope lies more than eleven standard errors away from zero – far too far to be a chance result.

Testing other hypothesized values

Question: A marketing study claims that each Euro of price costs *half* a lamp. Is this compatible with our data?

$$H_0 : \beta_1 = -0.5 \quad \text{vs.} \quad H_1 : \beta_1 \neq -0.5$$

Test

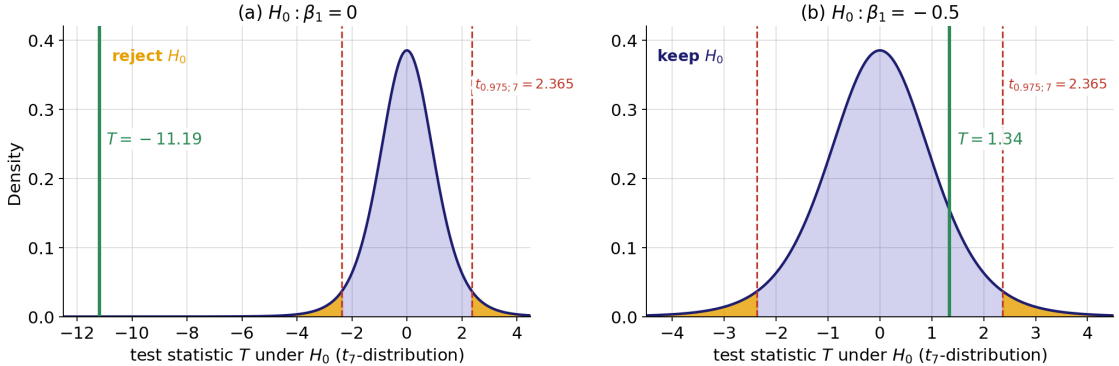
$$T = \frac{-0.4467 - (-0.5)}{0.03992} = \frac{0.0533}{0.03992} = 1.336, \qquad |T| = 1.336 < 2.3646 = t_{0.975;\,7}$$

$\Rightarrow$ **keep** H_0 $(p = 0.223)$.

Reading

The data cannot distinguish $\beta_1 = -0.5$ from the estimate -0.4467: the claim is *compatible* with the data – but not *proven*.

Two t-tests in one picture



Both tests use the same t_7-distribution and the same bars ± 2.365. (a) $T = -11.19$ lies far in the rejection region (orange); (b) $T = 1.34$ lies inside the acceptance region. The decision is nothing more than: **where does the statistic fall?**

One-sided test for the slope

Economic theory gives a direction

A price-sales function should *fall*. We want evidence for a negative slope:

$$H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_1 : \beta_1 < 0 \qquad \text{(left-sided)}$$

Lamp example

Reject if $T < -t_{1-\alpha,\, n-2} = -t_{0.95;\, 7} = -1.8946$:

$$T = -11.19 < -1.8946 \;\Rightarrow\; \text{reject } H_0, \qquad p = F_{t,7}(-11.19) = 0.000005.$$

As in Part 3

The direction must be fixed *before* looking at the data, and the one-sided p-value is half the two-sided one only if T points in the direction of H_1. R always reports the **two-sided** p-value.

Before you test – the assumptions checklist

1. **Linear model is correct** – residual plot without systematic pattern.
2. **Independent observations** – no autocorrelation (e.g. not the same store measured repeatedly).
3. **Constant variance** – no funnel in the residual plot.
4. **Normally distributed errors** – QQ-plot; for large n the central limit theorem makes the t-test approximately valid anyway.

Key takeaway

If the assumptions fail, the standard errors – and with them every t-test, confidence interval and p-value – are not reliable. The point estimates $\hat{\beta}_0, \hat{\beta}_1$ may still be fine.

Confidence interval for β_1

Construction – exactly as for μ in Part 3

$$\mathrm{CI}_{1-\alpha}(\beta_1) = \left[\hat{\beta}_1 - t_{1-\alpha/2,\,n-2}\ \mathrm{SE}(\hat{\beta}_1),\ \hat{\beta}_1 + t_{1-\alpha/2,\,n-2}\ \mathrm{SE}(\hat{\beta}_1)\right]$$

$$\mathrm{CI} = \Big[\ \underbrace{\hat{\beta}_1}_{\text{estimate}}\ \pm\ \underbrace{t_{1-\alpha/2,\,n-2}}_{\substack{\text{critical}\\ \text{value}}} \cdot\ \underbrace{\mathrm{SE}(\hat{\beta}_1)}_{\text{standard error}}\ \Big]$$

In words: For large n the critical value is close to $1.96 \approx 2$: $\hat{\beta}_1 \pm 2\ \mathrm{SE}(\hat{\beta}_1)$ is a quick 95 % interval.

Confidence interval – lamp example

Given: $\hat{\beta}_1 = -0.4467$, $\mathrm{SE}(\hat{\beta}_1) = 0.03992$, $t_{0.975;\,7} = 2.3646$.

$$\text{margin of error } e = 2.3646 \cdot 0.03992 = 0.0944$$

$$\mathrm{CI}_{0.95}(\beta_1) = [\,-0.4467 \pm 0.0944\,] = [\mathbf{-0.5411};\ \mathbf{-0.3523}]$$

Interpretation

The procedure covers the true slope in 95 % of all samples. Values of β_1 between -0.54 and -0.35 lamps per Euro are *compatible* with the data.

Duality with the test

$0 \notin \mathrm{CI} \Leftrightarrow H_0 : \beta_1 = 0$ is rejected; $\quad -0.5 \in \mathrm{CI} \Leftrightarrow H_0 : \beta_1 = -0.5$ is kept.

Confidence intervals – interpretation pitfalls

Careful with the interpretation

"With 95 % probability β_1 lies in *this particular* interval $[-0.5411; -0.3523]$" is, strictly speaking, **wrong**.

- β_1 is a fixed (unknown) number; the *bounds* of the interval are random.
- Correct: "The **procedure** yields, in 95 % of samples, an interval that covers β_1."

Intercept

The CI for β_0 is [45.15; 64.18]. It is computed in the same way – but β_0 (sales at price 0) has no substantive meaning here.

The p-value of a coefficient

Definition (as in Part 3)

The p-value is the probability, under $H_0 : \beta_1 = 0$, of a test statistic that is **at least as extreme** as the observed one:

$$p = P\big(|T| \geq |T_{\text{obs}}| \,\big|\, H_0\big) = 2\,F_{t,n-2}\big(-|T_{\text{obs}}|\big).$$

Hypothesis (lamp)	T	p (two-sided)	Decision at $\alpha = 0.05$
$H_0 : \beta_1 = 0$	-11.19	0.00001	reject
$H_0 : \beta_1 = -0.5$	1.34	0.223	keep

Key takeaway

$p < \alpha \Leftrightarrow |T| > t_{1-\alpha/2,\,n-2} \Leftrightarrow$ hypothesized value outside the CI.

Reading the regression output in R

```
> summary(lm(quantity ~ price))
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 54.66667    4.02519   13.58 2.76e-06 ***
price       -0.44667    0.03992  -11.19 1.02e-05 ***
---
Residual standard error: 1.546 on 7 degrees of freedom
Multiple R-squared:  0.947,  Adjusted R-squared:  0.9395
F-statistic: 125.2 on 1 and 7 DF,  p-value: 1.016e-05
```

`Estimate, Std. Error`	$\hat{\beta}_j$ and $\mathrm{SE}(\hat{\beta}_j)$
`t value, Pr(>|t|)`	T for $H_0 : \beta_j = 0$ and its two-sided p-value
`Residual standard error`	$\hat{\sigma} = \sqrt{\mathrm{RSS}/(n-2)}$ with $n - 2 = 7$ degrees of freedom
`Multiple R-squared`	R^2 (adjusted R^2: see multiple regression)
`F-statistic`	tests all slopes jointly; here $F = T^2 = (-11.19)^2 = 125.2$

Significant $\neq$ important

The p-value measures evidence, not size

With many observations even a tiny slope becomes highly significant. The size of the effect is read from $\hat{\beta}_1$ and its confidence interval.

Advertising data ($n = 200$): sales on TV budget

$\hat{\beta}_1 = 0.0475$ (sales in 1000 units, budget in 1000 \$), SE $= 0.0027$, $T = 17.67$, $p < 0.0001$, $\text{CI}_{0.95} = [0.0422; 0.0528]$.

- *Significance:* overwhelming evidence that TV advertising and sales are related.
- *Size:* 1000 \$ more TV budget goes along with 42 to 53 additional units sold – whether this pays off is a business question, not a statistical one.

Key takeaway

Report the estimate *and* its confidence interval – not just the p-value or the stars.

Two questions about a new x_0

At a new price x_0 the prediction is $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$. How precise is it? It depends on the question:

Confidence interval

for the **mean** sales $E(Y \mid x_0)$ of *all* stores with price x_0:

$$\hat{y}_0 \pm t_{1-\alpha/2,\, n-2}\, \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Prediction interval

for the sales Y_0 of **one** new store with price x_0:

$$\hat{y}_0 \pm t_{1-\alpha/2,\, n-2}\, \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Key takeaway

The "1+" adds the error u_0 of the single store: a prediction interval is **always wider** and never shrinks below $\pm t\,\hat{\sigma}$, however large n becomes.

Prediction – lamp example

Given: $\hat{\sigma} = 1.5461$, $n = 9$, $\bar{x} = 100$, $S_{xx} = 1500$, $t_{0.975;\,7} = 2.3646$, new price $x_0 = 110$: $\hat{y}_0 = 54.67 - 0.4467 \cdot 110 = 5.53$.

Calculation

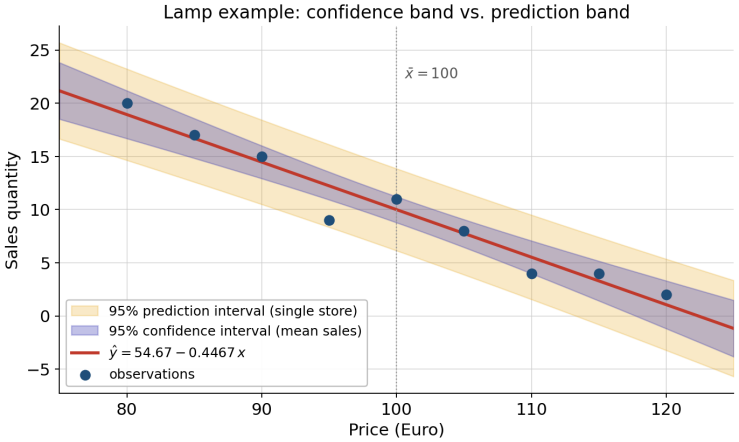
$$\text{CI: } 5.53 \pm 2.3646 \cdot 1.5461 \sqrt{\tfrac{1}{9} + \tfrac{100}{1500}} = 5.53 \pm 1.54 = [3.99;\ 7.07]$$

$$\text{PI: } 5.53 \pm 2.3646 \cdot 1.5461 \sqrt{1 + \tfrac{1}{9} + \tfrac{100}{1500}} = 5.53 \pm 3.97 = [1.57;\ 9.50]$$

Reading

At 110 Euro, stores sell on *average* between 4 and 7 lamps (95 % CI); a *single* new store will sell between about 2 and 9 lamps (95 % PI).

Confidence band vs. prediction band



- Both bands are narrowest at $\bar{x} = 100$ and widen towards the edges.
- The prediction band contains most of the points; the confidence band does not need to.

Extrapolation

At $x_0 = 125$ Euro the line predicts $\hat{y}_0 = -1.17$ lamps – impossible. Outside the data range the linear model breaks down.

Inference in simple regression with R (1/2)

Tests, *p*-values and confidence intervals

```
reg ← lm(quantity ∼ price)
summary(reg)                    # estimates, SE, t, p, R^2
confint(reg, level = 0.95)      # CI for beta0 and beta1
```

By hand

```
se_b1 ← summary(reg)$coefficients["price", "Std. Error"]
T ← (coef(reg)["price"] - (-0.5)) / se_b1    # H0: beta1 = -0.5
2 * pt(-abs(T), df = 7)                      # p-value 0.223
qt(0.975, df = 7)                            # critical value 2.3646
```

Inference in simple regression with R (2/2)

Prediction

```
new ← data.frame(price = 110)
predict(reg, new, interval = "confidence")     # mean sales: [3.99, 7.07]
predict(reg, new, interval = "prediction")     # one store:  [1.57, 9.50]
```

Note

`predict()` needs a data frame whose column has *the same name* as the explanatory variable in the model (here `price`).

Exercise – standard error, t and confidence interval

Problem statement

A simple linear regression on $n = 10$ observations gives $\hat{\beta}_1 = 2.5$, residual standard error $\hat{\sigma} = 4.0$ and $S_{xx} = \sum (x_i - \bar{x})^2 = 16$.

1. Compute the standard error $\mathrm{SE}(\hat{\beta}_1)$.
2. Compute the t-statistic for $H_0 : \beta_1 = 0$.
3. Compute the 95 % confidence interval for β_1.
4. Is the slope significant at the 5 % level? Justify in two ways.

Hint: $t_{0.975;\,8} = 2.306$.

Solution – standard error, t and confidence interval

$$\textbf{(1)} \quad \mathrm{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}} = \frac{4.0}{\sqrt{16}} = \mathbf{1.0}$$

$$\textbf{(2)} \qquad T = \frac{2.5 - 0}{1.0} = \mathbf{2.5}, \qquad \mathrm{df} = n - 2 = 8$$

$$\textbf{(3)} \qquad \mathrm{CI}_{0.95} = 2.5 \pm 2.306 \cdot 1.0 = [\mathbf{0.194}; \mathbf{4.806}]$$

(4) Decision

$|T| = 2.5 > 2.306$ – equivalently, 0 lies outside the CI. $H_0 : \beta_1 = 0$ is rejected at $\alpha = 0.05$ ($p \approx 0.037$).

In words: Common mistake: dividing $\hat{\sigma}$ by $\sqrt{n}$ – the standard error of the slope divides by $\sqrt{S_{xx}}$.

Exercise – inference for the shoe-size regression

Problem statement

For the body heights and shoe sizes ($n = 15$) we found $\hat{s} = 2.3147 + 0.2222\,k$. In addition: RSS $= 6.9432$, $S_{kk} = 2122.4$, $\bar{k} = 170.2$.

1. Compute $\hat{\sigma}$ and SE$(\hat{\beta}_1)$.
2. Test $H_0 : \beta_1 = 0$ at $\alpha = 0.05$.
3. Compute the 95 % confidence interval for β_1.
4. A rule of thumb says "one shoe size per 4 cm", i.e. $\beta_1 = 0.25$. Test this claim.
5. Give a 95 % prediction interval for the shoe size of a person with $k_0 = 180$ cm.

Hint: $t_{0.975;\,13} = 2.1604$.

Solution (1/3) – inference for the shoe-size regression

(1) Residual standard error and standard error

$$\hat{\sigma} = \sqrt{\frac{6.9432}{15-2}} = \sqrt{0.5341} = 0.7308, \qquad \mathrm{SE}(\hat{\beta}_1) = \frac{0.7308}{\sqrt{2122.4}} = \frac{0.7308}{46.069} = 0.01586$$

(2) t-test for $H_0 : \beta_1 = 0$

$$T = \frac{0.2222}{0.01586} = 14.01 \; > \; 2.1604 = t_{0.975;\,13} \quad \Rightarrow \quad \text{reject } H_0 \quad (p = 3.2 \cdot 10^{-9}).$$

In words: Body height is a highly significant predictor of shoe size.

Solution (2/3) – inference for the shoe-size regression

(3) Confidence interval

$$0.2222 \pm 2.1604 \cdot 0.01586 = 0.2222 \pm 0.0343 = [\mathbf{0.1879}; \mathbf{0.2565}]$$

Between 0.19 and 0.26 shoe sizes per cm are compatible with the data.

(4) $H_0 : \beta_1 = 0.25$

$$T = \frac{0.2222 - 0.25}{0.01586} = -1.75, \qquad |T| = 1.75 < 2.1604 \quad \Rightarrow \quad \text{keep } H_0 \ (p = 0.103).$$

Consistent with the CI: $0.25 \in [0.1879; 0.2565]$.

Careful

"Keep H_0" does not prove the rule of thumb – the data are merely *compatible* with it.

Solution (3/3) – inference for the shoe-size regression

(5) Prediction interval at $k_0 = 180$

$$\hat{s}_0 = 2.3147 + 0.2222 \cdot 180 = 42.31$$

$$42.31 \pm 2.1604 \cdot 0.7308 \sqrt{1 + \tfrac{1}{15} + \tfrac{(180-170.2)^2}{2122.4}} = 42.31 \pm 2.1604 \cdot 0.7308 \cdot 1.0546 = 42.31 \pm 1.66$$

$$\text{PI}_{0.95} = [\mathbf{40.65}; \mathbf{43.98}]$$

In words: A person of 180 cm will, with 95 % prediction probability, wear a shoe size between about 40.5 and 44. The CI for the *average* shoe size of all such persons is much narrower: [41.78; 42.84].

Exercise – used cars from A to Z

Problem statement

A dealer records the resale value y (in 1000 Euro) of six used cars against their age x (in years):

x_i (age)	1	2	3	4	5	6
y_i (value)	19	18	17	15	12	9

1. Estimate $\hat{\beta}_0, \hat{\beta}_1$ and write down the fitted line.
2. Compute the residuals, RSS, $\hat{\sigma}$ and R^2.
3. Compute $\mathrm{SE}(\hat{\beta}_1)$, the t-statistic and the 95 % confidence interval for β_1.
4. Predict the value of a 7-year-old car with a 95 % prediction interval.

Hint: $t_{0.975;\,4} = 2.776$.

Solution (1/3) – used cars from A to Z

Summaries: $n = 6$, $\bar{x} = 3.5$, $\bar{y} = 15$, $x_i - \bar{x} = (-2.5, -1.5, -0.5, 0.5, 1.5, 2.5)$, $y_i - \bar{y} = (4, 3, 2, 0, -3, -6)$.

$$S_{xy} = -35, \qquad S_{xx} = 17.5, \qquad S_{yy} = 74$$

(1) Coefficients and fitted line

$$\hat{\beta}_1 = \frac{-35}{17.5} = -\mathbf{2}, \qquad \hat{\beta}_0 = 15 - (-2) \cdot 3.5 = \mathbf{22}, \qquad \hat{y} = 22 - 2\,x$$

Each additional year of age lowers the value by 2000 Euro on average.

Solution (2/3) – used cars from A to Z

(2) Fit

Fitted values $\hat{y}_i = (20, 18, 16, 14, 12, 10)$, residuals $\hat{u}_i = (-1, 0, 1, 1, 0, -1)$:

$$\text{RSS} = 4, \qquad \hat{\sigma} = \sqrt{\tfrac{4}{6-2}} = 1.0, \qquad R^2 = 1 - \tfrac{4}{74} = 0.946$$

(3) Inference on the slope

$$\text{SE}(\hat{\beta}_1) = \frac{1.0}{\sqrt{17.5}} = 0.239, \qquad T = \frac{-2}{0.239} = -8.37, \qquad |T| > 2.776 \;\Rightarrow\; \text{reject } H_0$$

$$\text{CI}_{0.95} = -2 \pm 2.776 \cdot 0.239 = [-2.66;\ -1.34]$$

In words: The annual loss of value lies between 1340 and 2660 Euro; the interval excludes 0, in line with the t-test.

Solution (3/3) – used cars from A to Z

(4) Prediction at $x_0 = 7$

$\hat{y}_0 = 22 - 2 \cdot 7 = 8$ and $(x_0 - \bar{x})^2 = 12.25$:

$$8 \pm 2.776 \cdot 1.0\sqrt{1 + \tfrac{1}{6} + \tfrac{12.25}{17.5}} = 8 \pm 3.79 \;\Rightarrow\; \mathrm{PI}_{0.95} = [4.21;\, 11.79]$$

In words: The interval is wide: n is small, $x_0 = 7$ lies outside the observed ages (extrapolation), and it targets *one* car, not the average.

Common mistake

Dropping the "1+" under the root: 8 ± 2.58 is the confidence interval for the *average* 7-year-old car – far too narrow for a single car.

Overview

1 Relationship Between Two Metric Variables

2 Simple Linear Regression

3 Goodness of Fit: the Coefficient of Determination

4 Residual Analysis

5 Inference in Simple Regression

6 Multiple Linear Regression
- From one to many explanatory variables
- Interpreting the coefficients
- Inference for single coefficients
- F-test
- Goodness of fit and model choice
- Qualitative explanatory variables
- Interactions

7 Diagnostics in Multiple Regression

8 Summary

Why one explanatory variable is rarely enough

Intuition

Sales do not depend on the price alone – advertising, season, location and competitors matter as well. Real questions involve **several** explanatory variables at once.

- Multiple regression studies the effect of one variable **while holding the others fixed** (*ceteris paribus*).
- Separate simple regressions cannot do this: if the explanatory variables move together, each simple slope quietly absorbs the effects of the others.

Key takeaway

One joint model disentangles the effects – this is what makes multiple regression the workhorse of quantitative research.

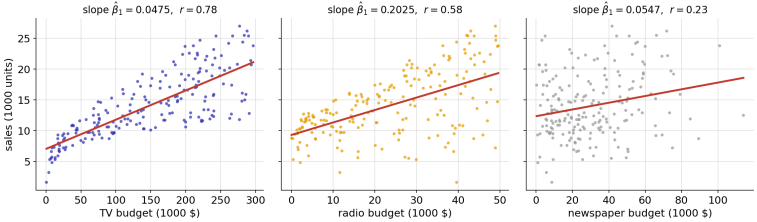
Our second story: the advertising data

The brief

A company has spent advertising budgets on TV, radio and newspaper in $n = 200$ markets and recorded the sales. *"Where should next year's budget go?"*

Variables

- Y: `sales` (in 1000 units)
- X_1, X_2, X_3: `TV`, `radio`, `newspaper` budgets (in 1000 $)



Source: Advertising data from James, Witten, Hastie, Tibshirani: An Introduction to Statistical Learning, www.statlearning.com

The multiple linear regression model

Model with p explanatory variables

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + u_i, \qquad i = 1, \ldots, n,$$

with the same four assumptions on the error terms u_i as before.

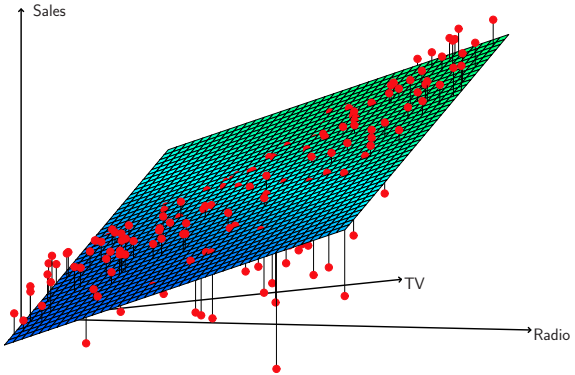
Interpretation of a coefficient

β_j is the expected change in Y when X_j increases by one unit **and all other explanatory variables are held fixed**.

Advertising

$$\texttt{sales}_i = \beta_0 + \beta_1 \, \texttt{TV}_i + \beta_2 \, \texttt{radio}_i + \beta_3 \, \texttt{newspaper}_i + u_i$$

Geometry: from a line to a plane



- With two explanatory variables the fitted model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ is a **plane**.
- The residuals are again the *vertical* distances of the points from the plane.
- With p variables: a "hyperplane" – no longer drawable, but the same idea.

Source: ISLR, Figure 3.5 (sales on TV and radio).

Least squares with several variables

Same principle

Choose $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$ such that the sum of squared residuals is minimal:

$$\min_{\hat{\beta}_0,\ldots,\hat{\beta}_p} \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}\right)^2.$$

- The solution is unique as long as no variable is an exact linear combination of the others.
- There is no simple "covariance over variance" formula any more – in matrix notation $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, which software computes for us.
- Residuals, RSS, TSS and $R^2 = 1 - \text{RSS}/\text{TSS}$ are defined exactly as before.

Multiple regression with R

Fitting the model

```
ad ← read.csv("../data/Advertising.csv", row.names = 1)
m3 ← lm(sales ∼ TV + radio + newspaper, data = ad)  # sales explained by ...
summary(m3)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.938889   0.311908   9.422   <2e-16 ***
TV           0.045765   0.001395  32.809   <2e-16 ***
radio        0.188530   0.008611  21.893   <2e-16 ***
newspaper   -0.001037   0.005871  -0.177     0.86
---
Residual standard error: 1.686 on 196 degrees of freedom
Multiple R-squared:  0.8972,  Adjusted R-squared:  0.8956
F-statistic: 570.3 on 3 and 196 DF,  p-value: < 2.2e-16
```

Interpreting the coefficients – ceteris paribus

Advertising (budgets in 1000 \$, sales in 1000 units)

- **TV:** $\hat{\beta}_1 = 0.0458$ – holding radio and newspaper fixed, 1000 \$ more TV budget goes along with about $0.0458 \cdot 1000 = 46$ additional units sold.
- **Radio:** $\hat{\beta}_2 = 0.1885$ – holding TV and newspaper fixed, 1000 \$ more radio budget goes along with about 189 additional units.
- **Newspaper:** $\hat{\beta}_3 = -0.0010$ – essentially zero once TV and radio are in the model.
- **Intercept:** $\hat{\beta}_0 = 2.94$ – expected sales without any advertising (2940 units).

The clause that matters

"Holding the others fixed" – a coefficient in a multiple regression is a *partial* effect, not the total association of X_j with Y.

Simple vs. multiple slopes

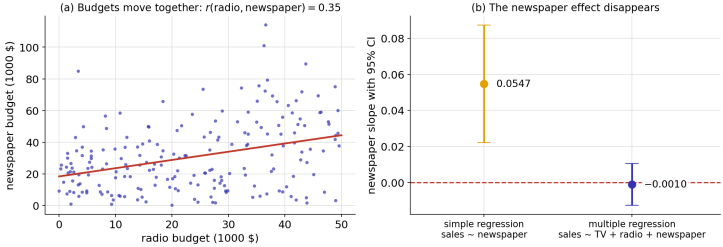
	Simple regression (one variable each)	**Multiple regression** (all three)
TV	0.0475	0.0458
Radio	0.2025	0.1885
Newspaper	0.0547 ($p = 0.001$)	−0.0010 ($p = 0.86$)

What just happened?

On its own, newspaper looks like a significant driver of sales – but its effect *disappears* once TV and radio are in the model.

In words: TV and radio hardly change – TV is almost uncorrelated with the other budgets, and radio is the variable that drives sales. Newspaper changes completely.

Confounding: why the newspaper slope collapses



- Markets with a large radio budget also spend more on newspapers ($r = 0.35$) – and radio really drives sales.
- In the simple regression, newspaper acts as a **proxy** for radio and "borrows" its effect.

Key takeaway

Confounding: a simple association can be produced by a third variable – include all relevant variables.

Exercise – why newspaper "works" alone

Problem statement

A simple regression of sales on the newspaper budget gives a positive, significant slope (0.0547, $p = 0.001$). In the multiple regression the newspaper coefficient is -0.0010 with $p = 0.86$. The correlation between the newspaper and radio budgets is 0.35.

1. Explain how newspaper can look useful on its own but useless in the joint model.
2. Which variable is the actual driver?
3. State the general lesson for interpreting simple correlations.

Solution – why newspaper "works" alone

1. **Confounding:** markets with high radio budgets also tend to have higher newspaper budgets. Radio raises sales. In the simple regression, the newspaper budget carries information about the radio budget that "travels with it" and inherits a positive slope. Once radio is included, radio absorbs this effect.
2. **Driver:** radio (and TV). Newspaper is a stand-in for radio in the simple regression.
3. **Lesson:** a simple association can be produced entirely by an omitted variable. To assess the effect of X_j *holding the others fixed*, all relevant variables must be in the model.

Common mistake

Concluding "newspaper advertising can never work". The model only says that newspaper adds nothing *given TV and radio, in these data* – an observational, not a causal statement.

t-test for a single coefficient

Same recipe – new degrees of freedom

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_1 : \beta_j \neq 0, \qquad T = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)} \sim t_{n-p-1} \text{ under } H_0$$

- $n - p - 1$ degrees of freedom: $p + 1$ coefficients were estimated (simple regression: $p = 1 \Rightarrow n - 2$).
- $\hat{\sigma} = \sqrt{\mathrm{RSS}/(n - p - 1)}$; the standard errors are computed by software.
- **Reading of H_0:** "X_j adds nothing *given the other variables in the model*".

Advertising ($n = 200$, $p = 3$, df $= 196$, $t_{0.975;\,196} = 1.972$)

Newspaper: $T = -0.0010/0.00587 = -0.18$, $|T| < 1.972 \Rightarrow$ keep H_0 ($p = 0.86$). TV ($T = 32.8$) and radio ($T = 21.9$): reject.

Confidence intervals in multiple regression

Construction

$$\mathrm{CI}_{1-\alpha}(\beta_j) = \hat{\beta}_j \pm t_{1-\alpha/2,\, n-p-1} \cdot \mathrm{SE}(\hat{\beta}_j)$$

Radio

$$0.18853 \pm 1.972 \cdot 0.00861 = 0.18853 \pm 0.01698 = [0.1715;\, 0.2055]$$

Holding TV and newspaper fixed, 1000 \$ more radio budget goes along with 172 to 206 additional units sold.

	TV	radio	newspaper
95 % CI	[0.0430; 0.0485]	[0.1715; 0.2055]	[−0.0126; 0.0105]

In words: The newspaper interval contains 0 – the same message as its t-test.

Is *any* explanatory variable useful? – the F-test

Hypotheses

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \quad \text{vs.} \quad H_1 : \text{at least one } \beta_j \neq 0$$

F-statistic

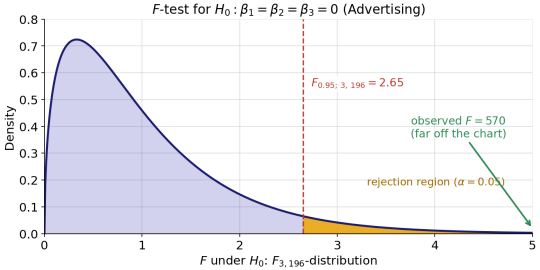
$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)} \sim F_{p,\, n-p-1} \text{ under } H_0$$

- Numerator: explained variation *per explanatory variable*.
- Denominator: unexplained variation per residual degree of freedom ($= \hat{\sigma}^2$).
- Large $F \Rightarrow$ the variables explain more than chance would $\Rightarrow$ reject H_0 (right-sided test).

F-test – advertising example

Given: TSS $= 5417.15$, RSS $= 556.83$, $n = 200$, $p = 3$.

$$F = \frac{(5417.15 - 556.83)/3}{556.83/196} = \frac{1620.11}{2.8410} = \mathbf{570.3} \gg 2.65 = F_{0.95;\,3,\,196}$$



In words: H_0 is rejected: at least one advertising medium is related to sales ($p < 2.2 \cdot 10^{-16}$).

Why not just several t-tests?

Multiple testing

With p explanatory variables and $\alpha = 0.05$ per test, even if *no* variable matters we expect about $0.05 \cdot p$ "significant" t-tests by chance alone – with $p = 20$ that is one false alarm per regression.

The F-test

One single test for all slopes together – its Type I error stays at α, however many variables are in the model. Consult it **first**, then look at the individual t-tests.

Special case

In simple regression ($p = 1$) the F-test and the two-sided t-test are identical: $F = T^2$ (lamp: $125.2 = (-11.19)^2$).

Exercise – the F-test

Problem statement

A multiple regression with $p = 3$ explanatory variables is fitted on $n = 100$ observations, giving $\text{TSS} = 1000$ and $\text{RSS} = 100$.

1. Compute R^2 and the F-statistic.
2. Compare with $F_{0.95;\,3,\,96} = 2.70$. What do you conclude?
3. Suppose instead 100 explanatory variables were all truly useless. About how many would appear "significant" at $\alpha = 0.05$ in individual t-tests?

Solution – the F-test

(1) R^2 and F

$$R^2 = 1 - \frac{100}{1000} = 0.9, \qquad F = \frac{(1000 - 100)/3}{100/(100 - 3 - 1)} = \frac{300}{1.0417} = \mathbf{288}$$

(2) $288 \gg 2.70 \Rightarrow$ reject $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$: at least one variable is related to Y.

(3) Each t-test rejects a true H_0 with probability 0.05, so about $100 \cdot 0.05 = 5$ variables would look significant purely by chance – which is why the F-test is consulted first.

Common mistake

Reading a significant F as "all three variables matter". It only says that *at least one* does; the t-tests show which.

R^2 and adjusted R^2

R^2 never decreases

Adding *any* variable – even pure noise – can only lower RSS, so R^2 never decreases. Comparing models by R^2 always favours the larger model.

Adjusted R^2

$$\bar{R}^2 = 1 - \frac{\mathrm{RSS}/(n-p-1)}{\mathrm{TSS}/(n-1)} = 1 - (1-R^2)\,\frac{n-1}{n-p-1}$$

Penalises each additional variable: $\bar{R}^2$ only rises if the new variable reduces RSS by more than chance would.

In words: The residual standard error $\hat{\sigma} = \sqrt{\mathrm{RSS}/(n-p-1)}$ is penalised in the same way.

Model choice – dropping newspaper

Model	$\hat{\beta}_{\text{TV}}$	$\hat{\beta}_{\text{radio}}$	R^2	$\bar{R}^2$	$\hat{\sigma}$
TV + radio + newspaper	0.0458	0.1885	0.89721	0.89564	1.6855
TV + radio	0.0458	0.1880	0.89719	0.89615	1.6814

Decision

The slopes barely move, R^2 is practically unchanged, $\bar{R}^2$ rises and $\hat{\sigma}$ falls: **keep the simpler model** `sales ~ TV + radio`.

Common mistake

Arguing with the unchanged R^2 – plain R^2 can only fall when a variable is dropped. Adjusted R^2 and $\hat{\sigma}$ are the fair judges.

Prediction in multiple regression

Same two intervals as before

For a market with TV $= 100$ and radio $= 20$ (in 1000 \$), the model `sales ~ TV + radio` predicts

$$\hat{y}_0 = 2.921 + 0.04575 \cdot 100 + 0.18799 \cdot 20 = 11.26 \text{ (1000 units)}.$$

	95 % interval	Question
Confidence interval	[10.99; 11.53]	mean sales of all such markets
Prediction interval	[7.93; 14.58]	sales of *one* such market

Key takeaway

With $n = 200$ the mean is estimated precisely – but a single market still varies by about $\pm 2\hat{\sigma} \approx \pm 3.3$.

What if an explanatory variable is categorical?

Many relevant variables are qualitative (student yes/no, region, brand) – but regression needs numbers.

Solution: dummy variable (0/1 coding)

$$D_i = \begin{cases} 1 & \text{if person } i \text{ is a student,} \\ 0 & \text{otherwise (baseline).} \end{cases}$$

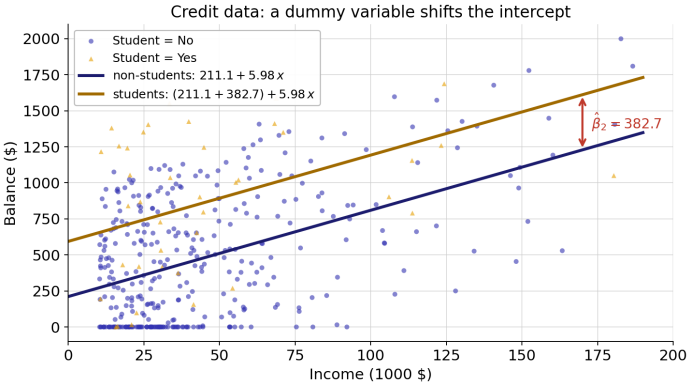
Credit data ($n = 400$): balance on student status

$$\widehat{\texttt{Balance}} = 480.37 + 396.46 \cdot D$$

Non-students: 480.37 \$ on average; students: $480.37 + 396.46 = 876.83$ \$. The coefficient of D is the **difference in means** between the groups ($p < 0.0001$).

Source: Credit data from James et al.: An Introduction to Statistical Learning.

A dummy shifts the intercept



$$\widehat{\texttt{Balance}} = 211.1 + 5.98\,\texttt{Income} + 382.7\,D$$

- Two **parallel** lines: same slope for income, intercept shifted by $\hat{\beta}_2 = 382.7$.
- Reading: at the *same* income, students carry on average 382.7 $ more balance.

K categories: use $K - 1$ dummies

Rule

A variable with K categories is coded with $K - 1$ dummies; the remaining category is the **baseline** represented by the intercept.

Why not K dummies? (dummy variable trap)

The K dummies always sum to 1 – exactly the column of the intercept. One variable would be an exact linear combination of the others, and least squares has no unique solution.

Credit data: region $\in$ {East, South, West}, baseline East

$$\widehat{\texttt{Balance}} = 531.00 - 12.50\, D_{\text{South}} - 18.69\, D_{\text{West}}$$

Both dummy coefficients are *differences to East* ($p = 0.83$ and $p = 0.77$): no evidence for regional differences.

Exercise – dummy coding

Problem statement

1. From $\widehat{\texttt{Balance}} = 480.37 + 396.46\, D_{\text{Student}}$, give the predicted mean balance of students and of non-students. Interpret 396.46.
2. From $\widehat{\texttt{Balance}} = 531.00 - 12.50\, D_{\text{South}} - 18.69\, D_{\text{West}}$ (baseline East), give the predicted mean balance in each region. How large is the difference between South and West?
3. Why are only $K - 1$ dummies used for a variable with K categories?

Solution – dummy coding

(1) Non-students ($D = 0$): 480.37 \$; students ($D = 1$): $480.37 + 396.46 = 876.83$ \$. The coefficient 396.46 is the **difference in average balance** between students and non-students.

(2) East $= 531.00$, South $= 531.00 - 12.50 = 518.50$, West $= 531.00 - 18.69 = 512.31$. South vs. West: $-12.50 - (-18.69) = +6.19$ \$.

(3) With K dummies plus intercept, the dummies sum to the intercept column (dummy variable trap) – the coefficients would not be identified. The dropped category becomes the baseline.

Common mistake

Reading -12.50 as "the mean of the South". Dummy coefficients are *contrasts* against the baseline, not group means.

When the effect of one variable depends on another

Interaction model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + u_i$$

The effect of x_1 is now $\beta_1 + \beta_3 x_2$ – it depends on x_2.

Advertising: `sales ~ TV * radio`

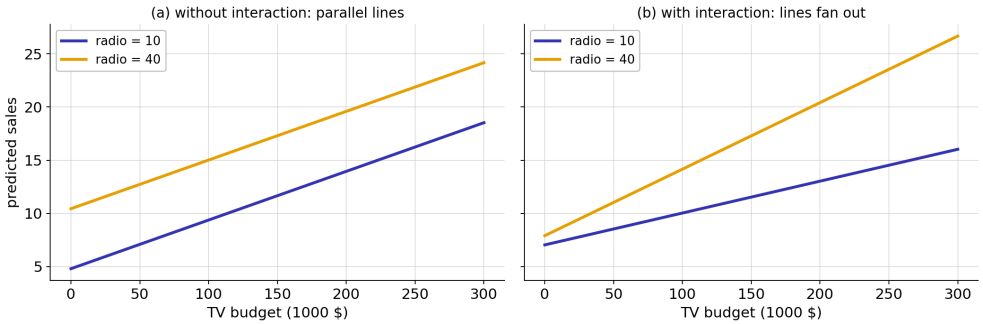
$$\widehat{\texttt{sales}} = 6.750 + 0.0191\,\texttt{TV} + 0.0289\,\texttt{radio} + 0.00109\,\texttt{TV} \cdot \texttt{radio}$$

Interaction: $T = 20.7$, $p < 0.0001$; R^2 rises from 0.897 to 0.968.

Hierarchy principle

If $x_1 x_2$ is in the model, keep x_1 and x_2 as well – even if their own t-tests are not significant.

Interaction, drawn: synergy between TV and radio



- Effect of 1000 \$ more TV budget: $0.0191 + 0.00109 \cdot \texttt{radio}$, i.e. 0.030 at radio $= 10$, but 0.063 at radio $= 40$.
- Parallel lines $=$ no interaction; lines that fan out $=$ synergy.

Overview

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4 Residual Analysis
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8 Summary

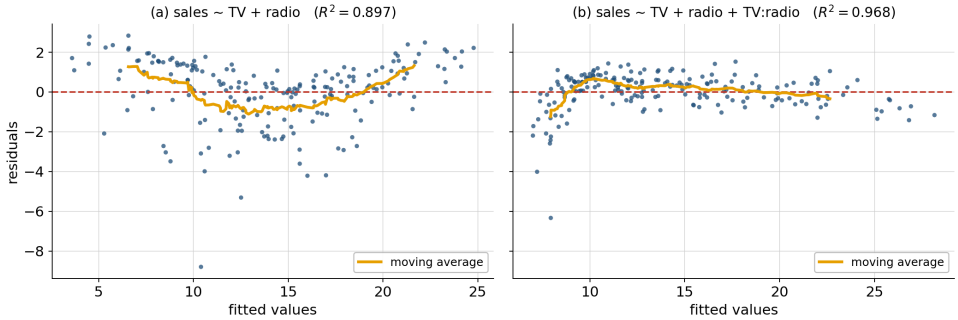
The classical problems – and how to spot them

Problem	Symptom	Tool
Non-linearity / missing term	curve in the residuals	residual plot
Heteroscedasticity	funnel in the residuals	residual plot
Non-normal errors	points off the line	QQ-plot
Outliers, high leverage	single points dominate	h_i, residuals
Collinearity	large SEs, unstable signs	correlations, VIF

Key takeaway

The residual plot and the QQ-plot of the simple regression work unchanged in multiple regression – plot the residuals against the fitted values $\hat{y}_i$.

Residual plot – advertising



- (a) The additive model leaves a **U-shaped** pattern: it over-predicts when one medium dominates and under-predicts when the budget is split.
- (b) With the interaction term the pattern largely disappears – the residuals *told* us that a term was missing.

Outliers and leverage in multiple regression

Outlier

A point with an unusually large residual $|\hat{u}_i|$ – unusual in y.

High-leverage point

A point with unusual values of the explanatory variables – unusual in x.

Rule of thumb for the leverage

The leverages h_i average $(p+1)/n$; flag points with

$$h_i > \frac{2(p+1)}{n} \qquad \text{(simple regression: } 4/n\text{)}.$$

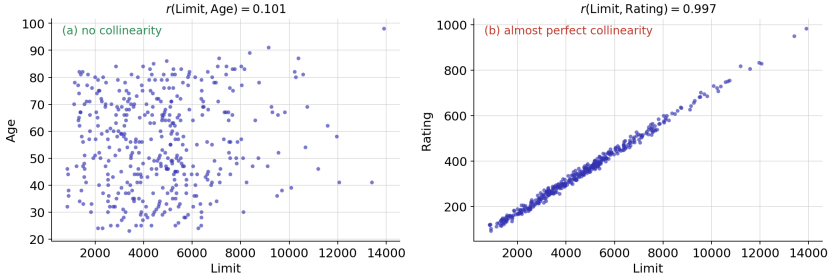
Worst case

A point that is *both* unusual in x and badly fitted in y can dominate the fit – inspect it, compare the fit with and without it.

Collinearity

Definition

Two or more explanatory variables are (almost) linearly dependent – they carry nearly the same information.



In words: When two variables move in lockstep (right), the data cannot separate their individual effects – holding one fixed while the other changes almost never happens.

Collinearity – consequences in the credit data

Model for `Balance`	$\hat{\beta}_{\text{Limit}}$	$\text{SE}(\hat{\beta}_{\text{Limit}})$	T	p
`Age + Limit`	0.1734	0.0050	34.5	< 0.0001
`Age + Limit + Rating`	0.0190	0.0630	0.30	0.76

Symptoms

Adding `Rating` (correlation 0.997 with `Limit`) multiplies the standard error of `Limit` by about 12.6 – the clearly relevant variable suddenly looks "insignificant".

In words: Collinearity does not spoil the *predictions* of the model; it spoils the *attribution* of the effect to individual variables.

Variance inflation factor (VIF)

Definition

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is the R^2 of a regression of x_j on all *other* explanatory variables.

- $\text{VIF}_j = 1$: no collinearity; $\text{VIF}_j > 5$ (some authors: 10): problematic.
- $\sqrt{\text{VIF}_j}$ is the factor by which collinearity inflates $\text{SE}(\hat{\beta}_j)$.
- Remedies: drop one of the redundant variables or combine them into one index.

Credit data (`Age`, `Limit`, `Rating`)

$\text{VIF}_{\text{Age}} = 1.01$, $\text{VIF}_{\text{Limit}} = 160.6$, $\text{VIF}_{\text{Rating}} = 160.7$ ($\sqrt{160.6} = 12.7$ – matching the SE ratio $0.0630/0.0050$).

Exercise – a diagnostics case study

Problem statement

House prices are regressed on `size`, `bedrooms`, age and `income` of the area ($n = 400$, $p = 4$). The diagnostics show:

- **Residual plot:** a clear U-shaped curve, and the spread of the residuals widens from left to right.
- **Leverage:** one point has $h_i = 0.30$.
- **VIF:** size 8.9, bedrooms 7.5, age 1.3, income 9.2.

Name every problem and suggest a remedy for each.

Solution – a diagnostics case study

1. **U-shape – non-linearity:** the linear model misses curvature. *Remedy:* add e.g. $\texttt{size}^2$ or transform a variable ($\log \texttt{size}$).
2. **Widening spread – heteroscedasticity:** standard errors, t- and F-tests are unreliable. *Remedy:* transform the response, e.g. $\log(\texttt{price})$.
3. **Leverage:** threshold $2(p+1)/n = 10/400 = 0.025$; $h_i = 0.30$ is twelve times as large. *Remedy:* inspect the point, compare the fit with and without it.
4. **Collinearity:** VIFs of 7.5–9.2 for size, bedrooms and income. *Remedy:* drop or combine redundant variables (e.g. keep size, drop bedrooms) and recompute the VIFs.

Common mistake

Deleting flagged points automatically. Diagnostics call for *inspection* and sensitivity checks – not for blind removal.

Multiple regression with R – overview

Fitting, testing, comparing

```
m3 ← lm(sales ∼ TV + radio + newspaper, data = ad)
summary(m3)              # t-tests, R^2, adjusted R^2, F-test
confint(m3)              # confidence intervals
m2 ← lm(sales ∼ TV + radio, data = ad)
anova(m2, m3)            # F-test: does newspaper add anything? (p = 0.86)
```

Dummies, interactions, diagnostics

```
lm(Balance ∼ Income + Student, data = cr)    # factor -> dummy automatically
lm(sales ∼ TV * radio, data = ad)            # TV + radio + TV:radio
plot(fitted(m2), resid(m2)); abline(h = 0)   # residual plot
hatvalues(m2)                                # leverages h_i
1 / (1 - summary(lm(Limit ∼ Age + Rating, data = cr))$r.squared)   # VIF
```

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Summary 1 – correlation and simple regression

Quantity	**Formula**
Covariance	$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$
Correlation coefficient	$r_{xy} = \frac{s_{xy}}{s_x\, s_y}, \qquad -1 \leq r_{xy} \leq 1$
Model	$y_i = \beta_0 + \beta_1 x_i + u_i, \qquad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$
Slope	$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{S_{xy}}{S_{xx}} = r_{xy}\frac{s_y}{s_x}$
Intercept	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
Coefficient of determination	$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$ $\left(= r_{xy}^2 \text{ in simple regression}\right)$

Summary 2 – model diagnostics

Residuals $\hat{u}_i = y_i - \hat{y}_i$

- **Residual plot** ($\hat{y}_i$ vs. $\hat{u}_i$): no pattern $\rightarrow$ model appropriate; curve $\rightarrow$ missing term; funnel $\rightarrow$ heteroscedasticity.
- **QQ-plot**: points on the line $\rightarrow$ normality plausible.

Influential points and collinearity

- Leverage $h_i > 2(p+1)/n$ (simple regression $4/n$): inspect, do not delete blindly.
- $\text{VIF}_j = 1/(1 - R_j^2) > 5$: collinearity inflates $\text{SE}(\hat{\beta}_j)$.

Coefficient of determination – assessment

$R^2 < 0.6$: weak $\quad 0.6 \leq R^2 < 0.8$: moderate $\quad R^2 \geq 0.8$: good $\quad$ (context-dependent!)

Summary 3 – inference for regression coefficients

Standard error and residual standard error

$$\hat{\sigma} = \sqrt{\frac{\text{RSS}}{n-p-1}}, \qquad \text{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}} \quad \text{(simple regression, } p = 1\text{)}$$

t-test, confidence interval, p-value

$$T = \frac{\hat{\beta}_j - \beta_{j,0}}{\text{SE}(\hat{\beta}_j)} \sim t_{n-p-1}, \qquad \hat{\beta}_j \pm t_{1-\alpha/2,\, n-p-1}\, \text{SE}(\hat{\beta}_j), \qquad p = 2\, F_{t,n-p-1}(-|T|)$$

Prediction at x_0 (simple regression)

$$\hat{y}_0 \pm t_{1-\alpha/2,\, n-2}\, \hat{\sigma} \sqrt{\underbrace{1}_{\text{PI only}} + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Summary 4 – multiple regression

- β_j: expected change in Y per unit of X_j, **holding all other variables fixed**.
- **Confounding**: simple and multiple slopes can differ completely (newspaper: $0.055 \to -0.001$).
- F-**test** for $H_0 : \beta_1 = \cdots = \beta_p = 0$: $F = \dfrac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)}$ – first check, then the t-tests.
- **Adjusted** $\bar{R}^2 = 1 - \dfrac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}$ for comparing models of different size.
- **Dummies**: K categories $\to K-1$ dummies; coefficients are differences to the baseline.
- **Interactions** $x_1 x_2$: the effect of x_1 depends on x_2; keep the main effects (hierarchy principle).

Summary 5 – important R commands

Description and estimation

```
plot(x, y)                  # scatter plot
cov(x, y); cor(x, y)        # covariance, corr.
reg ← lm(y ∼ x)             # linear model
coef(reg)                   # coefficients
abline(reg)                 # regression line
summary(reg)                # SE, t, p, R^2, F
```

Inference and diagnostics

```
confint(reg)                # CIs
predict(reg, new, interval = "prediction")
plot(fitted(reg), resid(reg))
qqnorm(resid(reg)); qqline(resid(reg))
hatvalues(reg)              # leverages
lm(y ∼ x1 + x2 + x3)        # multiple reg.
anova(small, large)         # F-test
```

Five classic pitfalls – final checklist

Forbidden sentence ×	**Correct version** ✓
"$r_{xy} = -0.97$, so price *causes* the sales to fall."	Correlation and regression show association, not causation.
"R^2 is high, so the model is correct."	Check the residual plot; a curved relationship can have a high R^2.
"The slope is not significant, so X has no effect."	No evidence found – the CI shows which effects are still compatible.
"In the multiple regression, $\hat{\beta}_j$ is the total effect of X_j."	It is the effect *holding the other variables fixed*.
"The model predicts -1.17 lamps at 125 Euro."	No extrapolation far beyond the observed x-range.

Outlook: Part 5 – Classification

What we have so far

Regression explains a **metric** variable Y (sales, shoe size, balance) by one or several explanatory variables – with full inference.

Next part

What if Y is **categorical** – default yes/no, spam or not? A straight line would predict "probabilities" below 0 or above 1. **Logistic regression** keeps the linear model on the log-odds scale and reuses everything from this part: coefficients, standard errors, tests, confidence intervals and confounding.

Key takeaway

Classification = regression for categories – plus the question of how to turn a probability into a decision.